MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY

INTELLIGENT OPERATIONS CENTER: A "SMART BRAIN" FOR SMART CITIES IN VIETNAM

by

NGUYEN NGOC HAI

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Abstract

Smart cities with the construction of Intelligent Operations Center (IOC) have been the key development of countries around the world, including Vietnam. Since cities are facing several problems, such as aging infrastructures, disparate data storage systems, the city authorities are unable to efficiently offer services sustainably, protect citizens or promote future economic growth without an integrated platform. In this thesis, we introduce an Intelligent Operations Center architecture which is developed on smart application systems with the focus on critical sectors such as city infrastructure, urban transit, the environment, economics and public opinion to address these issues in Vietnam. The IOC integrates monitoring, regulating, and other systems into a unified platform, allowing more efficient information delivery, better operational management, communication, cooperation, and decision-making. Being considered as the brain and central system of a smart city, we develop the IOC architecture with five main layers: (1) data integration layer, (2) storage layer, (3) data analytics layer, (4) data visualization layer and (5) application layer. Having implemented in more than 30 cities in Vietnam, the results highlight the efficiency and effectiveness of the IOC system not only in the decision-making process and city development planning but also in the economic development and increasing

living standards of citizens. Further research into the IOC system should focus on optimizing the system process and integrating Artificial Intelligence models in the application layer.

Acknowledgments

I would like to express my utmost appreciation to all those who made it possible for me to complete this thesis. I would like to thank the FPT School of Business and Technology, for providing me with the resources needed to undertake this project. I am deeply obliged to my supervisor, Dr. Truong Cong Doan, for his continuous support. This project could not have been completed without his encouragement and constant guidance.

Table of Contents

List of Tables

List of Figures

Chapter 1
Introduction

A smart city incorporates information and communication technologies into the planning, design, construction and operation of a city's infrastructure to address urban issues and builds a connected, technologically integrated and long-lasting infrastructure [1]. There are six important components of a smart city [2]: Smart Government includes solutions for efficiency and optimization of administrative units; Smart Economy emphasizes investment solutions – efficient production and flexible labor market; Smart Environment describes solutions for clean energy, low-energy consumption and smart building; Smart Living focuses on improving the living standard for residents in terms of healthcare, consumption, housing conditions and safety; Smart Mobility aims to improve the efficiency and quality of transportation, including cost-saving mobility solutions and emission reduction; Smart People towards an open and inclusive society to increase a city's or community's economy and innovation. Information regarding buildings, residents, gadgets and assets is analyzed for this purpose to effectively regulate urban flows via real-time reactions.

To enhance city operations, smart city development needs a powerful "brain" to simplify data silos, execute information exchange, convergence and integrate the physical and digital worlds. The development of an Intelligent Operations Center (IOC), to solve the problem of urbanization management is one of the most optimal solutions. IOC, which is constructed on smart application systems, focuses on critical sectors such as city infrastructure, urban transit, the environment, public security, economics, and public opinion. As the smart brain and central nervous system of a city, IOC provides an executive board to support municipal officials in gaining insight into all parts of the city [3]. IOC helps city departments exploit information and data systems to change the ways of operating, managing and promoting creativity and instrumentation to meet the needs of city residents, improve the efficiency of coordinating cross-agency resources and mitigate the effect of service and operations disruptions.

IOCs have been implemented in different regions and countries around the world, however, they are developed to solve city problems themselves. Rio de Janeiro, for example, had developed the Rio Operations Center to enable geologists, field operations and security to collaborate to improve emergency responsiveness, risk prevention and management [4]. Copenhagen in Denmark focuses on developing a laboratory for applications in the city with the collaboration of the residents, decision-makers and businesses in Copenhagen Solutions Lab [5]. Seoul Transport Operation & Information Service (TOPIS) operates and regulates the city's overall traffic with the purpose of minimizing severe traffic problems [6]. Such approaches will make these IOCs support effectively in a certain aspect, yet show some limitations, such as shared data has not been considered thoroughly, deployment components are stand-alone and single solution-oriented, with no emphasis on building the platform. Consequently, these applications are difficult for mutual exchange and cannot inherit existing systems.

In recent years, IOC has been deployed in provinces and cities in Vietnam. Depending on each city's characteristics, these IOCs have different functions and tasks, although they followed the same architectural foundation. The Intelligent Environment Operations Center in Hue, for example, was constructed to solve environmental and infrastructure problems [7]. Da Lat was one of the first cities to develop IOC in Vietnam that helped the city's authorities better administer and regulate social activities [8]. Each government agency focuses on their specialized data, such as the Ministry of Public Security developed an IOC for Residential Database, the Ministry of Finance built an IOC for the Insurance sector. The government also developed an IOC with the goal of connecting all these above systems in a single platform.

Cities in Vietnam are confronted with aging infrastructures, limited resources, shifting populations and growing dangers. Unshared data information limits the ability to coordinate and cooperate between sectors, as well as between the government and society, creating a cumbersome and slow system that is difficult to expand. In most cities in Vietnam, critical information is frequently stored in separate systems across multiple, separated divisions, making situation awareness difficult and coordinating agency activities challenging. A city may not offer services sustainably, protect residents or drive future economic growth without an integrated picture of events, incidents, or potential crises. Addressing these problems with purely traditional approaches is no longer effective.

In this thesis, we propose an IOC architecture that solves the mentioned limitations with six main principles: Putting citizens and businesses as a core; increasing the data effectiveness; prioritizing deployment of platform solutions; ensuring flexible scalability; providing high integration ability; assuring safety and security. This architecture has been deployed to the government and 35 provinces and cities in Vietnam, which initially brought positive results such as promoting the growth of the ICT Index, PAPI, more information is shared to citizens, data is more connected between departments. The IOC helps city authorities better monitor and manage city services by providing comprehensive information about city daily operations through centralized data management, especially during the COVID-19 period.

1.1 Methodology

1.2 Background

The Intelligent Operations Center incorporates measuring, controlling and other systems into a single platform and more efficiently provides information, improving operational management and contributing to collaboration, cooperation and decision-making [9]. Individual city agencies were unable to exchange information with other departments since they were focused only on their operations. The IOC helps to share information across departments and agencies, allowing them to cooperate, predict and respond to problems such as weather, natural disasters and crises.

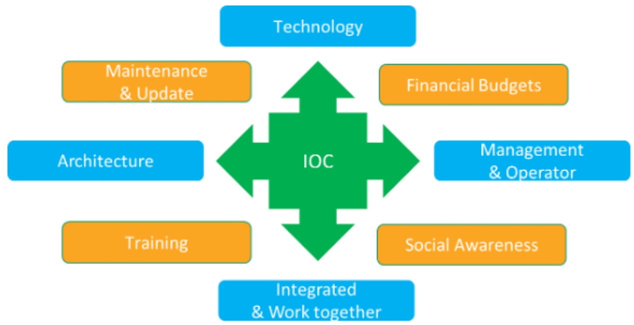

Fig. 1. Key components in developing an IOC system

There are some key components in the construction of an IOC, which interplay and are considered as standard in implementing technologies to develop the IOC as shown in Figure 1:

- Technology implementation depends on the city's existing architecture, accompanied by data traffic, servers, IDCs, operating procedures and human resources, which directly affect the reliability of the system.

- Financial budgets are one of the most concerned categories. The system will bring advantages to the city yet be considered from a cost perspective.

- Management & Operator: Close coordination among people, processes, policies and technology should be considered. Transforming gradually from a vertical (silo-type) model to a centralized one.

- Social Awareness: Training needs to be conducted for those involved in the system including authorities, business and citizens.

- Integrated & work together: Integrating different applications and technologies requires complex systems and highly qualified human resources. The state agencies, who are mainly responsible for management tasks, are not compatible with the capabilities required for human resources. Hence, these systems will be technically supported by providers.

- Training: The training process is carried out in transfer method (on the job training), so that city agencies can operate the system independently, then proceed to be in charge and connect future projects themselves.

- Maintenance and Update: The process of maintaining and upgrading the system should be invested in, to ensure the system works efficiently and keeps up to date with the latest technology.

- Architecture: The system is built based on the existing architecture and is suitable for oriented development. A completely new system can affect the integration and transformation of an ongoing one. As a result, it should be inherited from an existing architecture that easily develops and scales in the future.

1.3 IOC architecture

We propose an IOC architecture solution with the components and functions described in Figure 2.

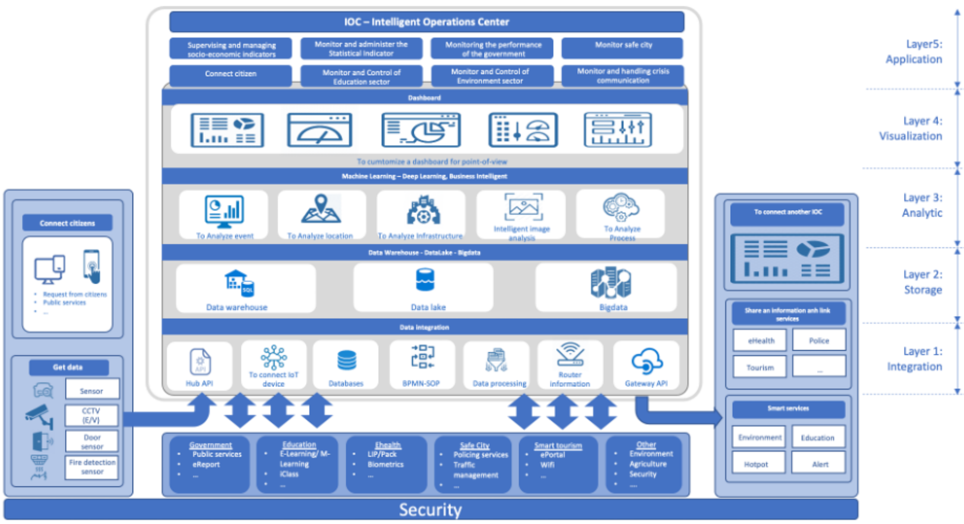

Fig. 2. Intelligent Operations Center Architecture

The IOC architecture includes five layers:

- Data integration layer: This layer gets the input data to the system and can forward notifications, alerts, KPI measurements to other systems. An Enterprise Service Bus (ESB) receives data from sources by different methods, it is also an infrastructure for consolidating services in the application environment [10]. Data integration lies in shared multi-dimensional integration through ESB where data are linked via protocol conversion.

- Storage layer: The data system is stored in this layer, including Data Warehouse – Data Lake – Big Data. These storages will be deployed based on stage and customer demands. The storage layer contains three main components: Specialized Database, Open Data and Data Warehouse.

- Specialized Database: The database is built by units within a city themselves and can be transformed into a shared database and open database

- Open Data: The shared database, such as Geographic Information System (GIS) database [11], is used for search and display information purposes. City citizens can freely get access to and share the data but must ensure the origin of such data is preserved. Open Data is particularly important for smart cities as citizens can view the operational status and access information easily.

- Data Warehouse: Different types of data are gathered and collected from city departments via data replication and synchronization to serve the needs of data sharing between units. Such component helps to connect and to build integrated services for citizens and businesses but requires strong authentication processes as sensitive data can be stored in the shared database.

- Data analytics layer: This layer connects to the storage layer, which is used to analyze the centralized data in the system, creating new values for the end-users.

- Data visualization layer: Analyzed data will be presented and visualized by dashboards and charts which show the overall status of a city. The dashboard helps city executives to obtain crucial insight into current challenges as it captures data from different sources, such as traffic reports, environment conditions and citizen satisfaction indices, and turns it into usable data. It also includes Standard Operating Procedures (SOPs) [12], situation reports and activity data that city authorities can evaluate properly and assess the situation for further improvements.

- Application layer: The arrogated, analyzed data from underneath layers are monitored and displayed in this layer. The city authorities need to prepare for both foreseen and unexpected situations, such as public health problems, traffic challenges. This will help city authorities in the decision-making process by providing ad-hoc reporting patterns and insights.

- Security layer: The security layer runs throughout the system. It covers other components in the system to ensure data security and information safety from the physical layers to the application layers, standards, policies and procedures. Information security systems operate on strategic policies, frameworks and implementation principles that are compliant with ISO 27001 [13].

1.4 IOC architecture characteristics

Three main characteristics of this IOC architecture are connection, sharing and data analysis that are expressed in the following advantages:

- The infrastructure platform utilizes Cloud Computing technology with flexible scalability that can be shared across departments to optimize investment and operating costs.
- Data is centralized from the IoT database systems to form an open Data Warehouse. A centralized information-sharing platform optimizes costs for storage systems, serves the

needs of sharing and analyzes data which facilitate the development of integrated and intelligent services.

- This architecture can help city agencies and departments integrate information from different systems to develop an interactive, connected platform that encourages collaboration, improves efficiency and promotes better decision-making. The city authorities can be kept informed, address public concerns, plan for events and respond to unanticipated emergencies.

1.5 Technology used

The IOC system is a combination of diverse and complex technologies, provided by technology corporations after analysis, comparison and testing. The IOC architecture is developed on shared computing infrastructure; a multi-cloud model is used accordingly to maximize the system security. The architecture is developed following Software-as-a-Service (SaaS) model on the Cloud Foundry platform [14] combined with the Zenko open-source code [15].

Table 1. Major components in IOC Architecture

No	**Layer**	**Technology**
1	Data integration	Based on the open-source Apache Nifi and Apache Sqoop, Pentaho Data integration [16], along with Oracle Data integration, we build the data integration system at different levels for different types of users in the system.
2	Data Warehouse, Data Lake, Big Data	We customize the system to meet the requirements of the city by collaborating with Cloudera [17].
3	Data analytics	We create a set of tools to analyze data by combining the systems on the Cloudera and Apache Lucene platform.
4	Data visualization	We use Power BI – Microsoft [18] for data visualization.

Chapter 2 System architecture

The IOC is an extensive system with many different components; therefore, the IOC architecture will be divided into different layers and components depending on the perspective. In which, 02 main views will be:

- List of components (modules) in the IOC system
- List of classes and methods of communication in the IOC system

In this chapter, the document will describe these two perspectives to clarify the IOC system architecture.

2.1 IOC's modules

Based on the IOC architecture framework in chapter 1, the IOC will include the following components.:

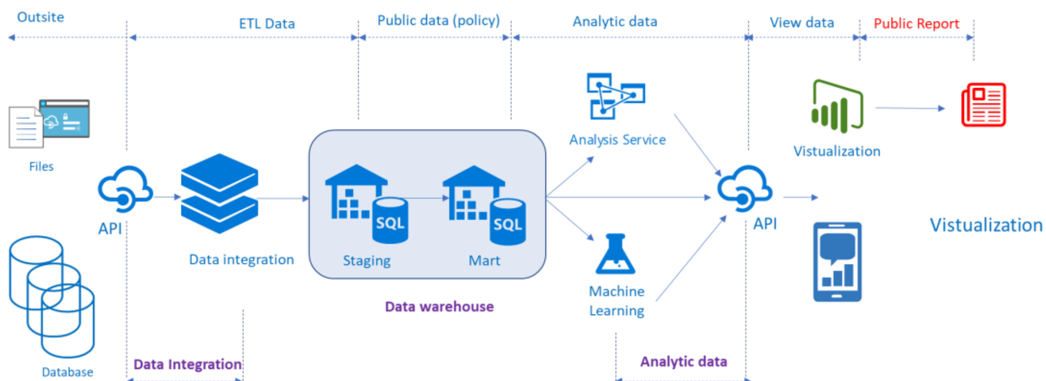

Fig. 2.1**.** Module of Intelligent Operations Center

The system consists of 3 main components

- Data Integration: data integration.
  - This component connects to existing domains according to the method that the domain offers (API, Webserver, Database, Files, ...)
  - This is an actively connected system; the IOC service provider side will bear the system load
  - Depending on the frequency of the data provider, the IOC system will synchronize and set the corresponding frequency to retrieve data
  - The whole module is divided into 2 phases: design and schedule to run automatically
- Datawarehouse: data warehouse
  - This component is used to hold the entire structured data of the system - Staging
  - This data will be classified by subject and presented to the service layer - Mart
- Data Analytics: this component serves to analyze data and build BI models from Datawarehouse

2.1.1 Data Integration

2.1.1.1 Introduction

- Data integration is taking data from different systems/sources into meaningful and valuable information. Integrate collected data in such a way that it can produce reliable, comprehensive, current, and accurate information for business analysis and reports.
- It developed from the open-source software Pentaho 2004. Module Data integration has been upgraded and modified to match the development environment of the overall software in the system.
- The Data Integration module is bundled into a package called BI Suite, which has five main modules:
- Module Data Integration Reporting: create reports; you can create notifications directly on the Browser by dragging and dropping columns or rows or measures from metadata or cubes.
- Data Integration Dashboard Module: create dashboards
- Data Integration (DI) Module: The most potent ETL tool used to aggregate, transform, and process data. While ETL tools are used most often in data warehouse environments, Data integration can also use DI for other purposes:
    - Move data between applications or databases
    - Export data from database to a flat-file (file)
    - Load data into the database
    - Clean up data
    - Integration of applications
- Data Integration OLAP module: multi-dimension based data analysis (tool Schema Workbench)
- Data Integration Data Mining Module: is a set of tools for machine learning and data mining

2.1.1.2 Mode of Operation

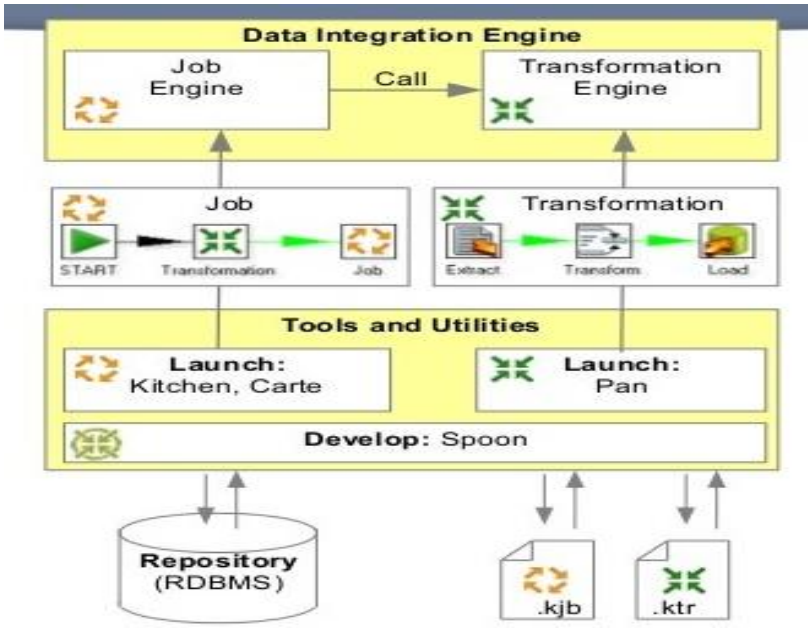

Fig 2.1.1 - Data Integration Architecture

- Job Engine and Transformation are 02 principal components used to integrate data from different sources through Develop- Spoon Interface.
- Job Engine: executes cron jobs based on Scheduler Services running in the background in the system.
- Transformation Engine: executes work as specified by the end-user. When transformation Engine combines with Scheduler Services and Par-amateur Services, we will have Job Engine.
    - Repository – RDBMS: general store information of the system
    - Kjb file and ktr file are 02 system definitions used to link the Tool for the end-user and the Data Integration Engine system (these files are defined by the end-user and pushed (Push) to the server through Responsibility Services
    - Job and Transformation are 02 components defined so that users can monitor, track and control execution steps in the system under a graphical interface.

2.1.2 Data Warehouse

2.1.2.1 Introduction

Data Warehouse is a concept first introduced in 1988 by two IBM researchers, Barry Devlin and Paul Murphy. This is where data is stored by an organization's electronic device to support data analysis and reporting. Today, the concept of DWH is understood in a broader context to include a set of methods, techniques, and technologies that can combine and support each other to collect and manage data from many different sources and environments to provide information to users.

Inmon, 2004 defines a data warehouse as a subject-oriented collection, integrated over time and persistently stored for decision making.

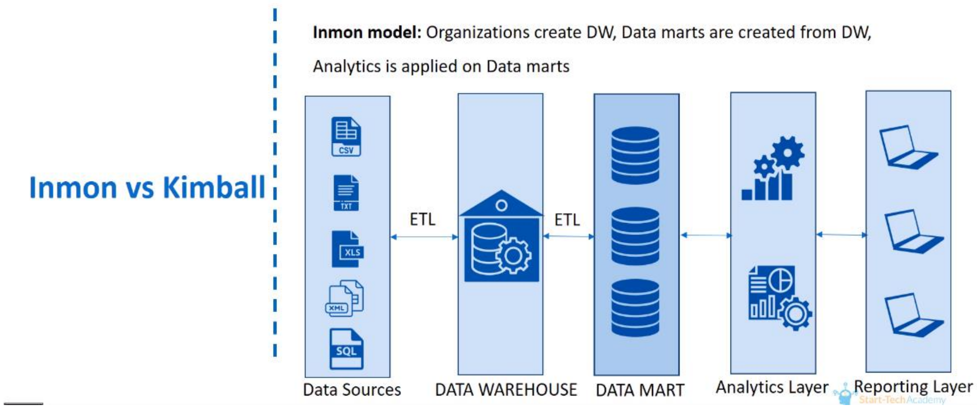

Fig 2.1.2-1 – Data warehouse concept according to Inmon, 2004

Kimball, 2005 defines a data warehouse as a system for extracting, cleansing, transforming, and loading data into a multidimensional data store to support query execution and decision making.

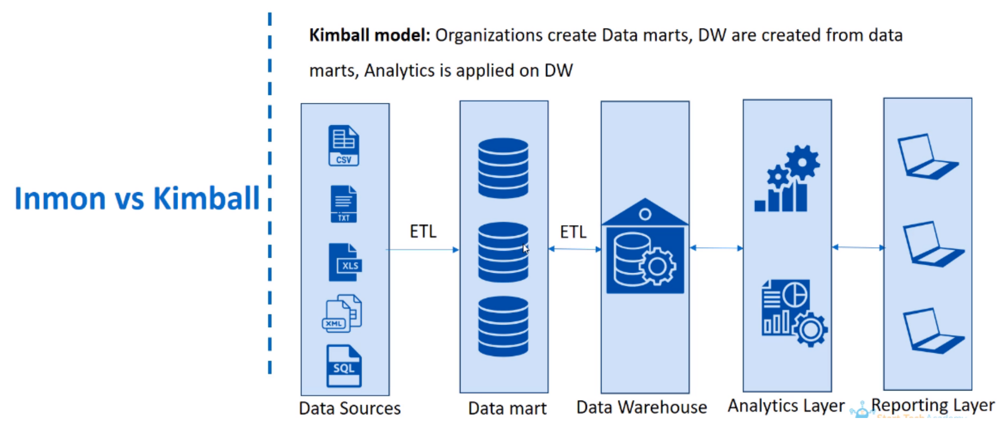

Fig 2.1.2-2 - Data warehouse concept according to Kimball, 2005

A data warehouse is a system that periodically retrieves and consolidates data from data sources into a standardized or multidimensional data container. It often keeps track of past data for many years and is queried for business intelligence or other analytical activities. It is updated in batches, not every time a transaction occurs in the source data.

Features of the data warehouse:

o Subject-oriented data is organized according to specific topics, containing only information relevant to decision support; provide a more holistic view of the organization

o Integrity: data coming from multiple sources

o Nonvolatile: data must be consistent over time

Time-varying: data used to detect trends, deviations, and long-term relationships for forecasting and comparison. Time is an important characteristic (daily, weekly, monthly, ...)

DataMart includes two entities: the target table (Fact Table) and the category table (Dimension Table).

o Fact Table contains measurable data

o Dimension Table has business description information, the properties of Dimension Table are usually text data and are descriptive; There is one primary key connected to the Fact Table.

The star schema is the most straightforward data warehouse schema in data warehousing, a star-like shape, with points emanating from one center; Centers are Fact Tables., output points are Dimension Tables.

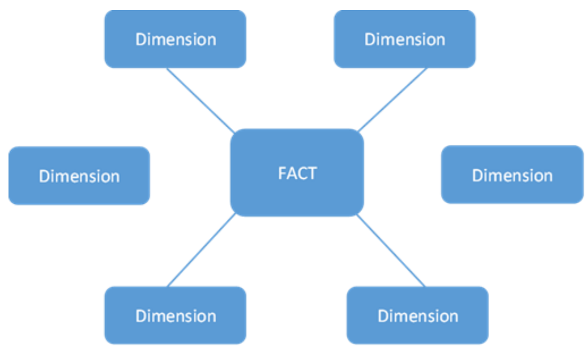

Fig 2.1.2-3: Star *schema*

A snowflake schema is a complex data schema that normalized Dimension Tables to reduce duplication, like snowflakes, whose points continue to grow (increase dimensions)

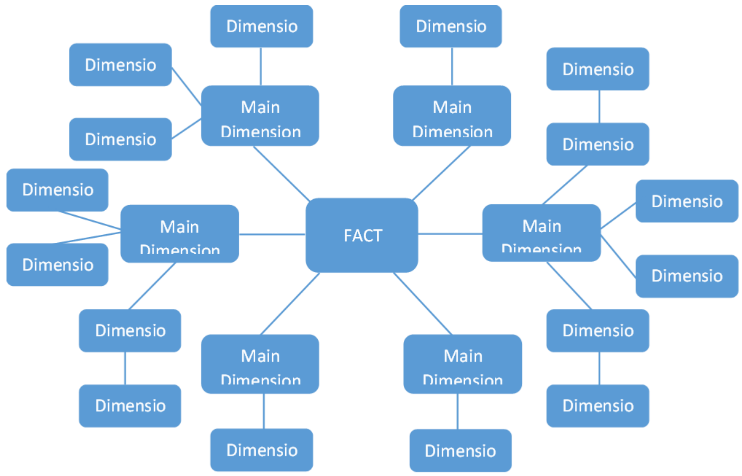

Fig 2.1.2-4: S*nowflake schema*

Change types update data when the source data changes.

- Type 1 (Type 1-Slowly Changing Dimension): overwrites the changed data into the Dimension table. This type 1 applies to source data that changes in the form of corrections or the updated part is not essential, does not change the meaning of the fact table.
- Type 2 (Type 2-Slowly Changing Dimension) allows for tracking the changes occurring in the dimension table and correctly linking the fact record to the dimension record that is in effect at the time of the fact record. When the DWH receives the updated source data, the system updates the old record state and adds a new record to the dimension table instead of overwriting it. This type 2 best represents data change over time because every entity change on the source data is recorded in the data warehouse.
- Type 3 (Type 3-Slowly Changing Dimension) is used when the dimension value changes, but DWH users can choose to use the new or old value; this type 3 is rarely used. The choice of which type of change to apply to a dimension entity depends on the business requirements of that dimension table.

2.1.2.2 Data warehouse design

The process of building a data warehouse

Fig 2.1.2-5: The process of building a data warehouse

Process explanation

Code	Step name	Explanation of steps

B01	ETL data into the Staging . layer	Use Transformations to load data from an external source into the DWH's Staging layer.
B02	Standardized data	Eliminate redundant, duplicate data at the Stag-ing layer according to business rules
B03.01	ETL Data Mart layer (Dimensional)	Transform data catalogs (computational dimensions) at the Staging layer into Dimension tables corresponding to the industry domain
B03.02	ETL DataMart (Fact)	Transforming the calculation criteria at the Staging layer into Fact tables corresponding to the industry field
B03.03	Building connection components	Build Procedures (Functions) for special handling: string processing, pre-calculation. Build, configure Job to check whether jobs are executed in a specific time & error warning (if any)

Table 2.1.2-1: The process of building a data warehouse

Staging stores snapshot data of the source system with the same structure as the tables in the source and with the addition of a DateTime field (NOW_CAPNHAT)

Data is taken from 2 primary data sources: units using HIS products and services being deployed and teams using other providers' IT services:

- API to get data from DB DataGuard products and sector’s services
- API receives data from the health data portal of the Ministry of Health

The list of data sources (APIs) declared and stored in the DB includes information about URL, body, content, method, table_name_stg, category_api corresponding to each module/field (category_api)

```sql
SELECT * FROM tgg_db_stg.etl_dm_api where category_api in ('HISTGG','HISPORTAL');
```

id_key	stt_api	url_api	body_api	content_api	method_api	table_name_stg	category_api	ngay_capnhat
10	0	http://egw.vnptasxh.vn/api/token/take	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	etl_api_token	HISTGG	2021-01-04 16:44:45
11	1	http://egw.vnptasxh.vn/api/ioc/dmBenhVien	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_dm_benhvien	HISTGG	2021-01-04 16:44:45
12	2	http://egw.vnptasxh.vn/api/ioc/dmKhoa	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_dm_khoa	HISTGG	2021-01-04 16:44:45
13	3	http://egw.vnptasxh.vn/api/ioc/thongKeThe...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_theo_doituong	HISTGG	2021-01-04 16:44:45
14	4	http://egw.vnptasxh.vn/api/ioc/thongKeChiPhi	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_chiphi	HISTGG	2021-01-04 16:44:45
15	5	http://egw.vnptasxh.vn/api/ioc/soLuotBenh...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_sobn_vaora	HISTGG	2021-01-04 16:44:45
16	6	http://egw.vnptasxh.vn/api/ioc/soBNTheoL...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_sobn_theo_loaihinh	HISTGG	2021-01-04 16:44:45
17	7	http://egw.vnptasxh.vn/api/ioc/soCaCapCu...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_soca_capcuu	HISTGG	2021-01-04 16:44:45
18	8	http://egw.vnptasxh.vn/api/ioc/soCaCapCu...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_soca_tuvong	HISTGG	2021-01-04 16:44:45
19	9	http://egw.vnptasxh.vn/api/ioc/soHoSoKCB	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_sohs_kcb	HISTGG	2021-01-04 16:44:45
20	10	http://egw.vnptasxh.vn/api/ioc/thongKeTai...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_tainan	HISTGG	2021-01-04 16:44:45
21	11	http://egw.vnptasxh.vn/api/ioc/thongKeThe...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_theo_tuoi	HISTGG	2021-01-04 16:44:45
22	12	http://egw.vnptasxh.vn/api/ioc/thongKeThe...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	his_tgg_tke_theo_khoa	HISTGG	2021-01-04 16:44:45
24	1	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_dm_benhvien	HISPORTAL	2021-01-08 16:57:30
25	2	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_dm_khoa	HISPORTAL	2021-01-08 16:57:30
26	3	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_tke_songay_dieutri	HISPORTAL	2021-01-08 16:57:30
27	4	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_tke_chiphi	HISPORTAL	2021-01-08 16:57:30
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33	10	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_tke_tainan	HISPORTAL	2021-01-08 16:57:30
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35	12	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_tke_theo_khoa	HISPORTAL	2021-01-08 16:57:30
38	13	https://congdulieuyte.vn/hPortal/rest/api/io...	{"user_name":"IOC_ADMIN","pass_word"...	application/json	POST	ehealth_tke_noitru	HISPORTAL	2021-01-11 15:48:43
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Fig 2.1.2-6: Medical Data Source API Data and Structure

Details of tables and structures (e.g., medical field – to see appendix)

2.1.3 Data Analytic:

2.1.3.1 Introduction

According to the Magic Quadrant 2017 regular report of Gartner's famous consulting firm, which evaluates and ranks data analytics and BI solution providers (pictured below), only three vendors appear in the quadrant. "Leaders" – Microsoft, Qlik, and Tableau are the top vendors. In which Microsoft and Tableau create a sizeable leading gap compared to Qlik and the rest of the vendors.

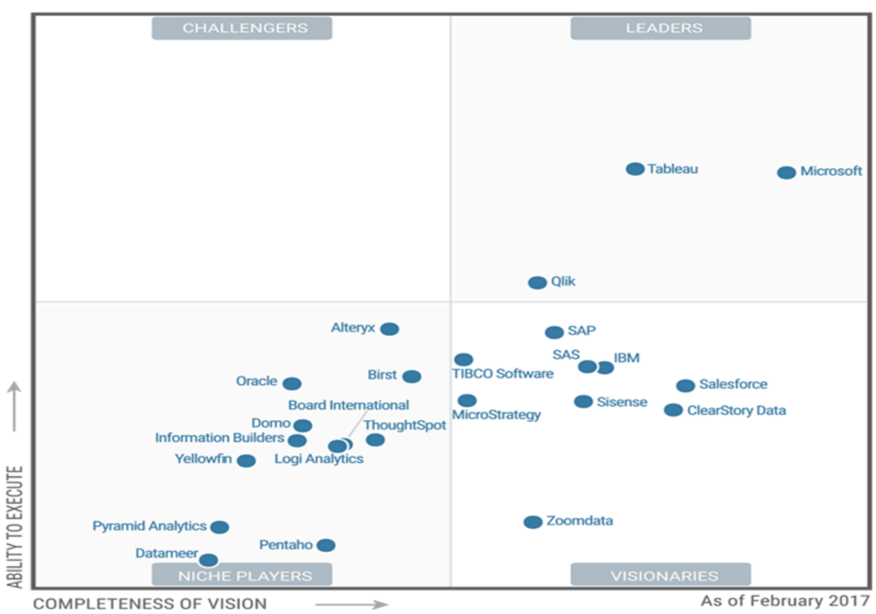

Fig 2.1.3-1: Magic Quadrant 2017 report

Compared to the previous year's report (2016) results, we can see that Microsoft and Tableau have widened the gap and affirmed their leading position.

Microsoft's leading Business Intelligence solution is Power BI, a solution powered on the Azure cloud platform, which also belongs to Microsoft. Individual users can also use an on-premises version of Power BI installed on their computers.

The main advantage of Microsoft is its relatively low total cost of ownership. The Desktop version for individuals is provided for free, while the cloud version for businesses has a relatively low monthly fee ($9.99/month/user).

Use the Microsoft Power BI tool to present data directly from DWH's DataMart tier in various charts, tables, graphs, and more,…

2.1.3.2 Mode of Operation

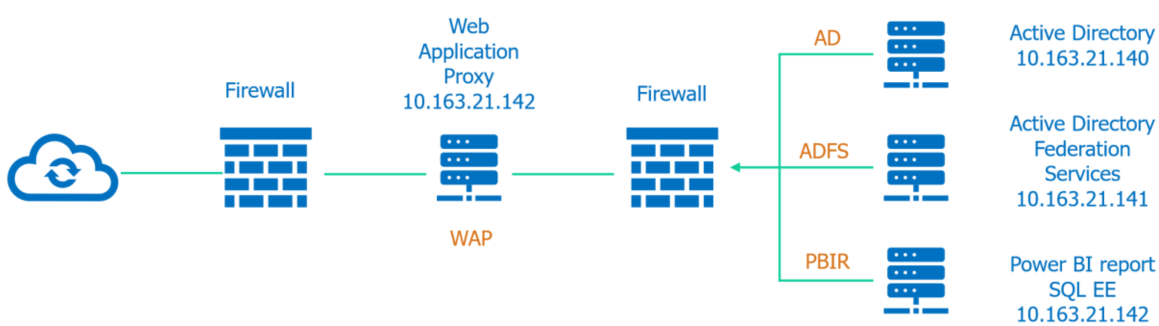

Fig 2.1.3-2: Power BI on-premise

This module is used according to the On-premise model. All components are deployed on the system infrastructure, not using the cloud model. In there:

- Active Directory và Active Directory Federation Services: serve user authentication

- Power BI report: The server contains the system's report file set

- Web Application Proxy: the server acts as a communication channel with the components inside the system

2.1.3.3 KPI

Based on KPIs in each field, the system will conduct detailed analysis and data visualization to help leaders have an overview of the data of the all sector in the province, specifically as follows: after:

– Medical examination and treatment field: the system needs to show detailed information about the number of medical examination and treatment visits by day/month, the number of visits to and from the hospital/medical facility, the number of outpatient/inpatient treatment visits. Residence, chart to monitor accident and injury situation, medical examination and treatment costs using health insurance, statistical chart of health insurance beneficiaries, number of records with health insurance/without insurance health insurance, examination records are male/female patients, follow up on what ages the patients come to the clinic, the most common and joint diseases, from which to conduct performance analysis such as a group of medical facilities. The medical facility has the highest number of medical examination/treatment visits, the most common group of diseases, the group of medical facilities with the highest rate of health insurance examination, the group of medical facilities receiving the most traffic accidents, etc.

- Monitoring the situation of Covid-19: Infectious diseases: the system needs to display the total number of Covid-19 cases, in which the number of patients in the community, the number of instances on entry, showing the number of points on the map by each province/city, region The chart shows the number of new topics by day in detail, the analysis of the infection situation by the time axis, the study of the province/city with the most points in the country.,…
- Public service sector: the system should show an overview of the processing of documents, administrative procedures over the months of the year, the percentage (%) of timely processing of documents, the total number of follow-up records. how much is received, online or face-to-face, the completion of the processing, tracking of overdue processed forms, pending applications, which areas have not been processed or are due, and from there, analyze the processing performance of records (which fields receive the most applications, which lots have many unprocessed or overdue forms, ...)
- In the field of organizational documents: the system needs to show information about the system usage, processing, and approval of documents (incoming, outgoing), from there to analyze the performance of multiple processing units. The most records, which units process the least, which teams use the least or most infrequently, etc.,…

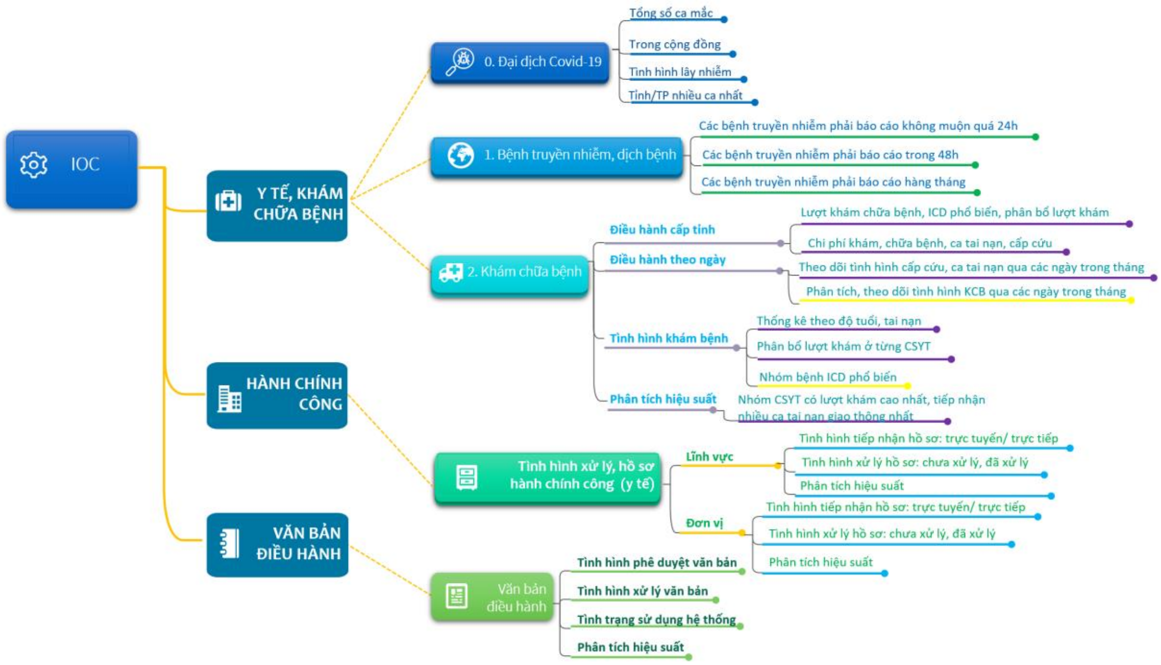

Fig 2.3.1-3: Operation's KPI

These KPIs will be the output of the system. However, with the dynamic design characteristics, these KPIs can be supplemented and adjusted to best suit the province/city without rebuilding the system.

2.2 IOC's component system

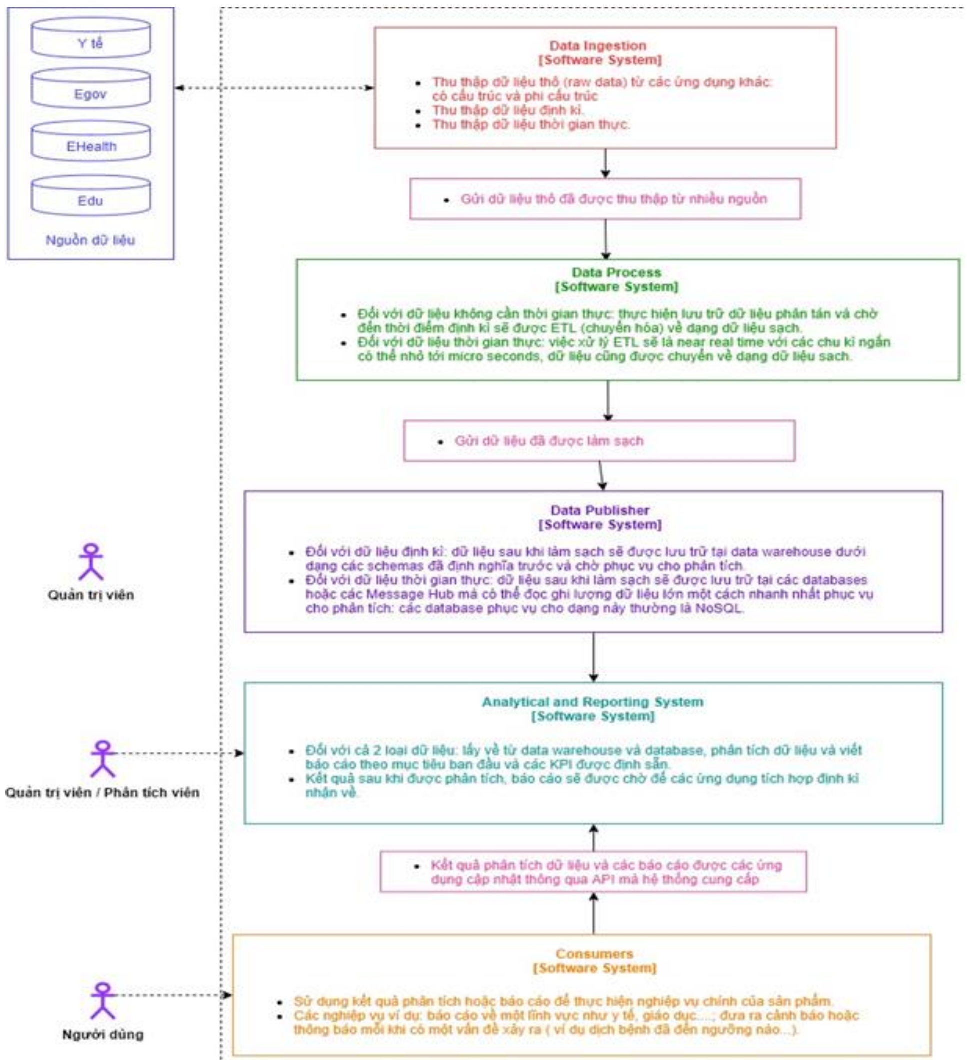

Fig 2.2: Part of IOC

Base on IOC's architechture on chapter 1, IOC's system has 04 part:

- Data Ingestion: This is a component that interacts directly with specialized data sources. This part is built as a hub that receives data directly from sources and temporarily saves it to 1 storage through different protocols like RestApi, DB Connection...
- Data Process: This is the component responsible for data processing and cleaning (ETL).
- Data Publisher: after being ETL according to the original data usage goal, the data is moved to the data warehouse and other storage designed to provide high-performance data querying for reporting and analysis.
- Analytical and Reporting System: This component is mainly responsible for creating data models that BA can derive and use for analysis and prediction.
- Consumer: Visualization and Alert System: it is a place to visualize analysis results as well as reports, warning over thresholds by a dashboard with charts, graphs, warnings.

2.2.1 Classtification and communication architecture

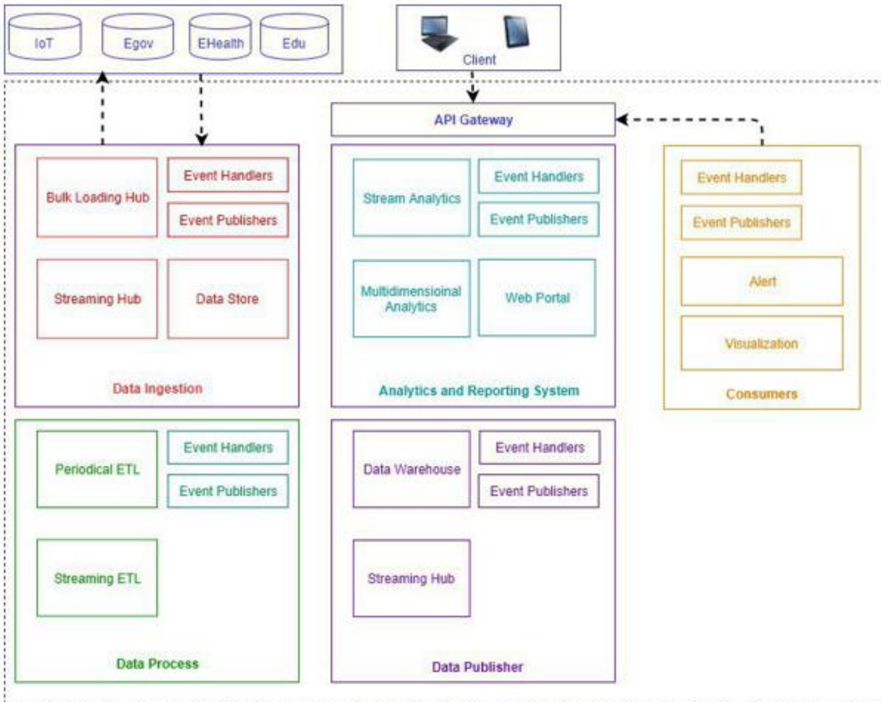

Fig 2.2.1-1:Class architecture of IOC

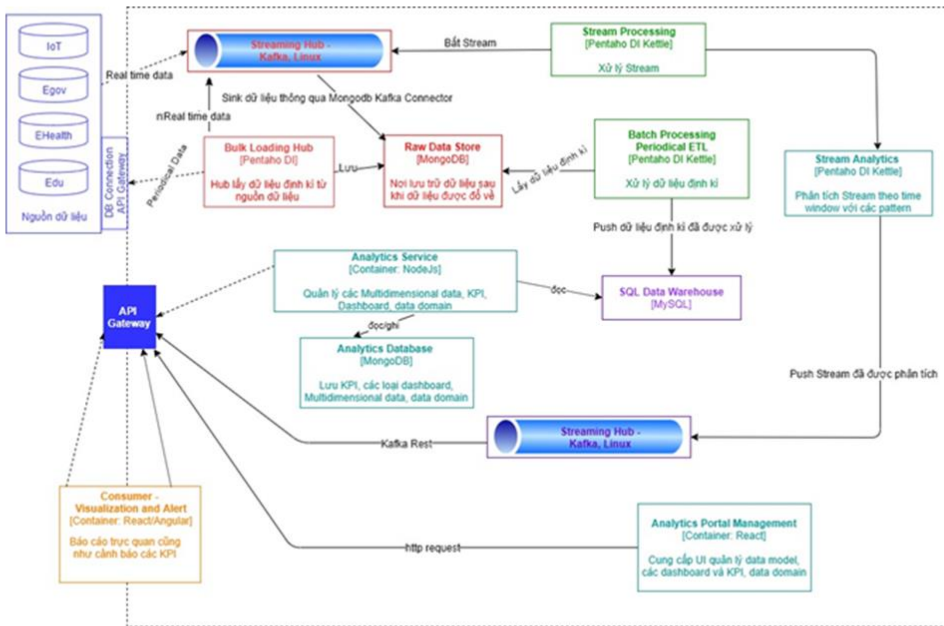

Fig 2.2.1-2: Communcation architecture of IOC

The entire system solution architecture is divided into separate components as follows:

– Streaming Hub is a component implemented and developed by the Apache Kafka streaming system. This component is responsible for receiving real-time data from external real-time data sources such as logs, IoT, etc. It is also the main communication channel between real-time data and the IOC system. as a place to store real-time data waiting to be analyzed.
– Bulk Loading Hub: component implemented by Pentaho Data Integration. This component is responsible for interacting with external API Gateways or Databases to get data periodically. Some data with a frequency close to real-time (non-real-time), with a significant update period (above seconds, minutes, hours) will be queued and allowed to connect.
– Raw Data Store: This component is a place to store data after it is downloaded from hubs temporarily. Datastore uses NoSQL MongoDB as storage for data types from streaming to batch.
– Stream Processing: component developed by Pentaho Data Integration Kettle. It is responsible for capturing data on the streaming hub and conducting stream processing (real-time ETL), waiting for analysis next step.
– Stream Analytics: component developed by Data Integration Kettle. Is the place responsible for analyzing streaming data after it has been pre-processed at Stream

Processing? After the analysis is complete, Stream analytics will transfer it to Streaming Hub to serve warnings and display data.

- Batch Processing: component developed by Pentaho Data Integration Kettle. This component aims to get the data dumped at the Data Store, conduct ETL processing of these periodic data, and save it to the Data Warehouse.
- SQL Data Warehouse: this is a place to store data periodically after it has been ETL, deployed by the MySQL server system.
- Analytics Service: it was built and developed by the NodeJS programming language. This component has primary responsibilities, including inputting data from sources and generating data models, from which data models serve different business problems.
- Analytics Portal Management: this is a component that plays the role of UI, the main channel of interaction with users to manage information of Data Models, display dashboards, and KPIs.
- API Gateway: This component aims to group APIs in the system into a single endpoint for easy access. In addition, this component supports caching, bypass, and retry mechanisms. This component is built and deployed by the Netflix open-source application.
- Consumer – Visualization and Alert: component built and developed by React language, is the place to get data after it has been analyzed, this data will be displayed visually by graphs, maps, charts; Along with that, there are warnings about problems with the set KPIs. This component can be designed within the IOC system or from the side of other external integration services.

2.2.2 Details of components in the system

In this section, the document will go into a detailed description of the child components (component) to form a separate part (container) described above.

2.2.2.1 Stream Processing:

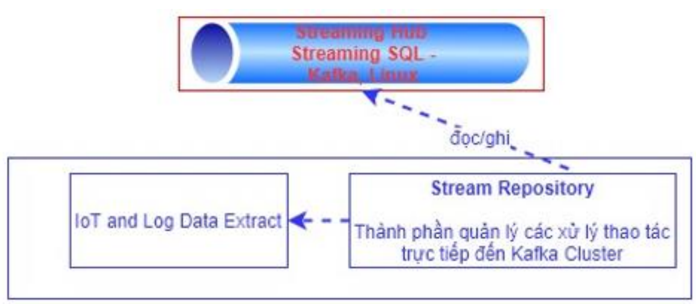

Fig 2.2.2-1: Stream processing

There is 2 main software to build; stream Processing includes:

- IoT and Log Data Extract: this is the main component of the processing flow, responsible for cleaning and transferring data online before analysis.

– Stream Repository: this is the component responsible for processing the task directly to the Kafka Cluster: catching the streams that are not cleaned up for processing.

2.2.2.2 Batch Processing:

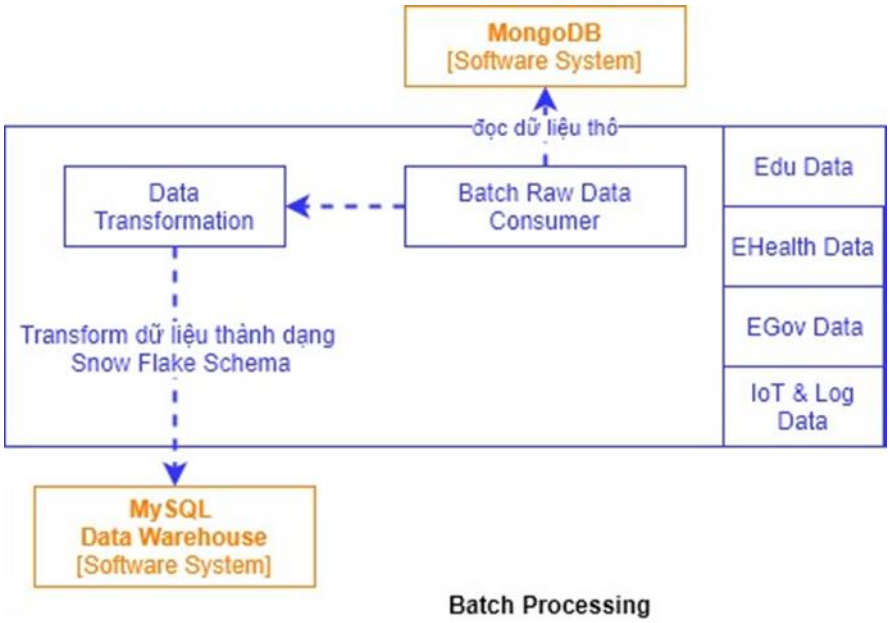

Fig 2.2.2-2: Batch processing

There are two main components to building Batch Processing, including:

– Batch Raw Data Consumer: this is the component responsible for reading the raw data stored after being pushed from the data hub.

– Data Transformation is the main component performing the cleaning, transforming data into Snow Flake Schema form and storing it in Data Warehouse to serve the analysis of large amounts of data..

2.2.2.3 Stream Analytics:

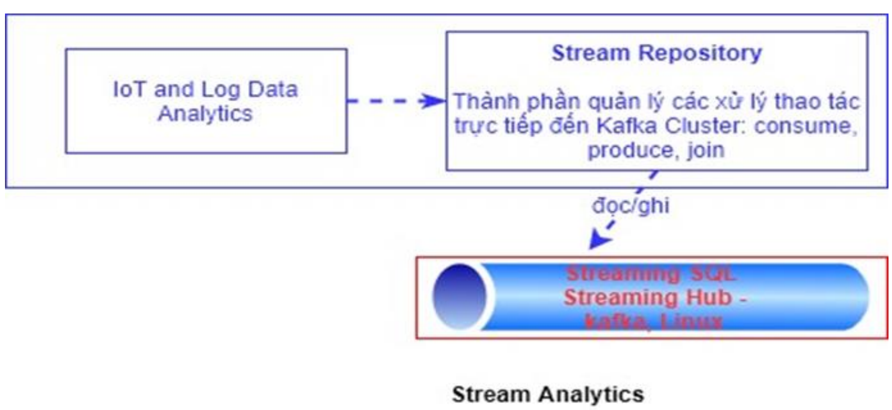

Fig 2.2.2-3: Stream analytics

Two main components build Stream Analytics, including:

- Stream Repository: this is the component responsible for handling operations directly to the Kafka Cluster: push the analyzed streams to Kafka for other services.
- IoT and Log Data Analytics: This main component analyzes streaming data according to patterns to give warnings about changes in data flow over time.

2.2.2.4 Analytics Service:

The main components to build Analytics Service:

Rest API: This is the component that provides a set of APIs in the form of Restful for other applications to access the information of the analyzed data. The piece is built and developed by the NodeJs programming language. Access to this API cluster requires two elements:

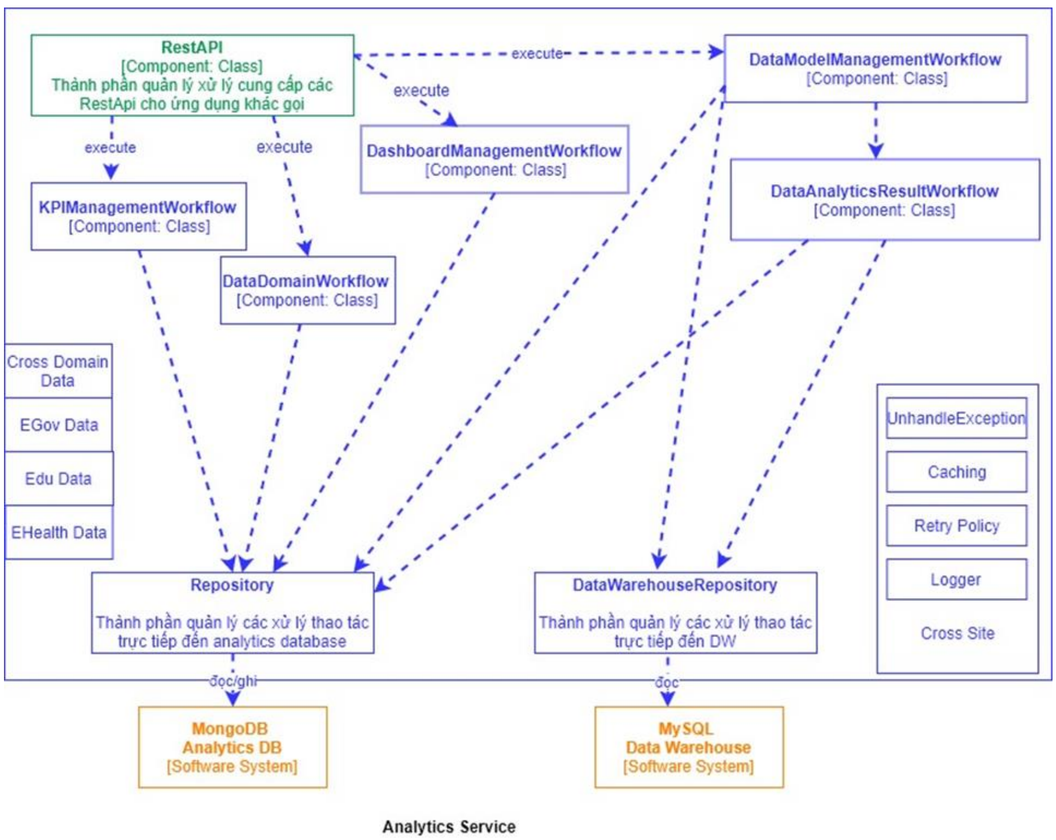

Fig 2.2.2-4: Analytics services

- Token: A string used to authenticate and authenticate the application's access to the API. The value of the token line is passed through the HTTP Header with the name "token."
- Data transmitted in JSON structure. Notes when designing these Rest APIs:
- Version: APIs must be defined and versioned. This makes it easy to upgrade the API.
- HTTP Method: Use the HTTP Method to describe the functionality of the API. We have 4 HTTP Methods, including getting, POST, PUT, DELETE, respectively, for four types: Get data, add data, update data, and delete data.

- HTTP Response Status Code: Use the HTTP Status code to describe the return status of the API. It is recommended that the API must correctly use the following status codes:
    - 2XX: It describes APIs with successful processing.
    - 4XX: It describes APIs with end-to-end error handling. Often we have the following errors:
    - 400 Bad Request: This Itrepresentss the mistake entered by the user.
    - 401 Unauthorized: error because the user has not authenticated to access.
    - 403 Forbidden: This error signals that the user does not have sufficient access rights.
    - 5XX: Error caused by the server, caused by the system.
- KPIManagementWorkflow: this component is responsible for managing the KPIs set by the user to ensure that the data system of that province runs well according to the set goals..

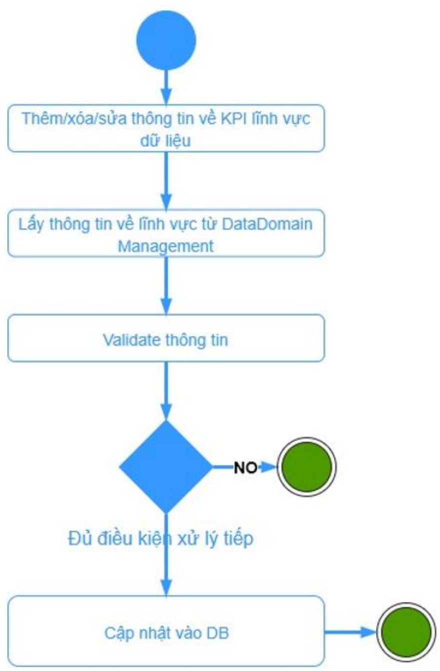

Fig 2.2.2-5: KPI Management workflow

DataDomainWorkflow: This component manages data domains in industries: healthcare, education, e-government, etc... It allows the system to select, add, edit, delete every time there is a new industry, allowing users to choose the sector to track the analysis data.

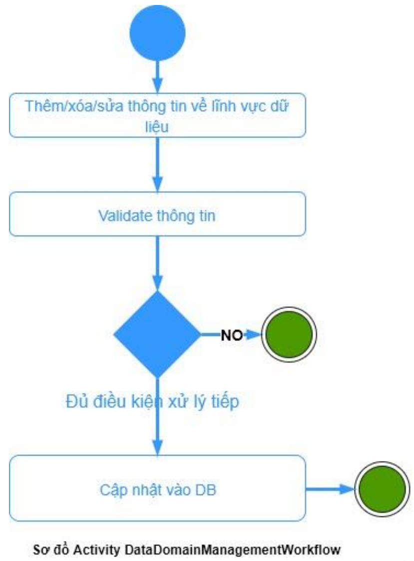

Fig 2.2.2-6: Activity Data Domain Management Workflow

DashboardManagementWorkflow: Components to manage the types of graphs and charts displayed on the dashboard for each field.

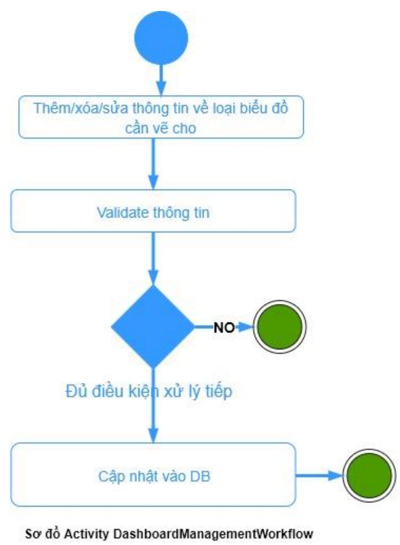

Fig 2.2.2-7: Dashboard Management Workflow

DataModelManagementWorkflow: the component responsible for managing data models with facts and dimension tables.

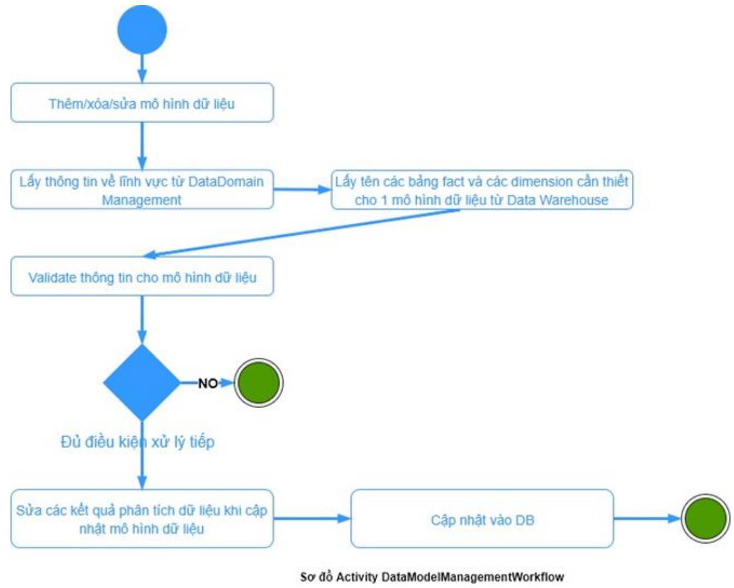

Fig 2.2.2-8: Data Model Management Workflow

DataAnalyticsResultWorkflow:

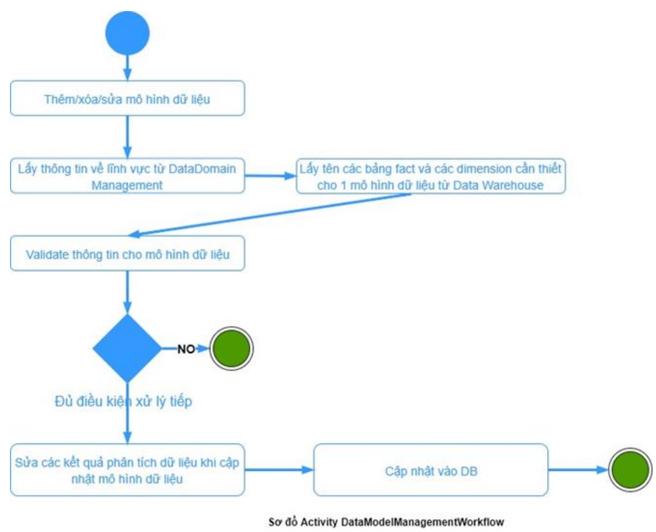

Fig 2.2.2-9: Data Analytics Result Workflow diagram

component responsible for changing the analysis data as the data model changes as well as the data changing periodically, called by DataModelManagementWork-flow.

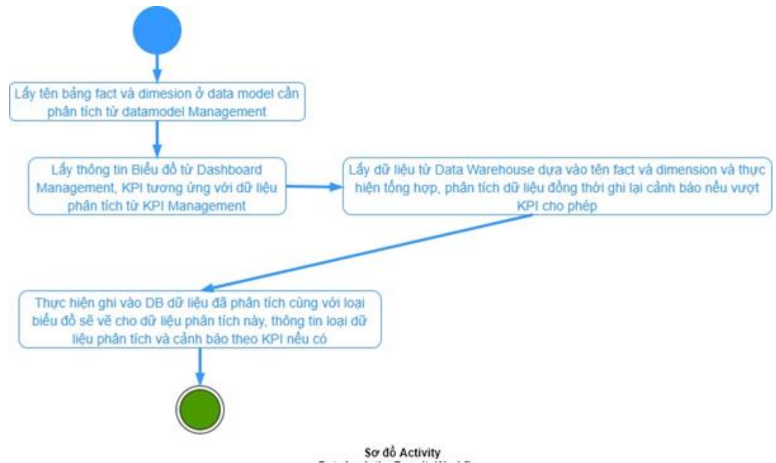

Fig 2.2.2-10: Activy diagram

DataWarehouseRepository: the component that manages Data Warehouse direct manipulation processes.

Repository: The component that manages the processing operations directly to the analytics database.

2.2.2.5 Visualization and alerts:

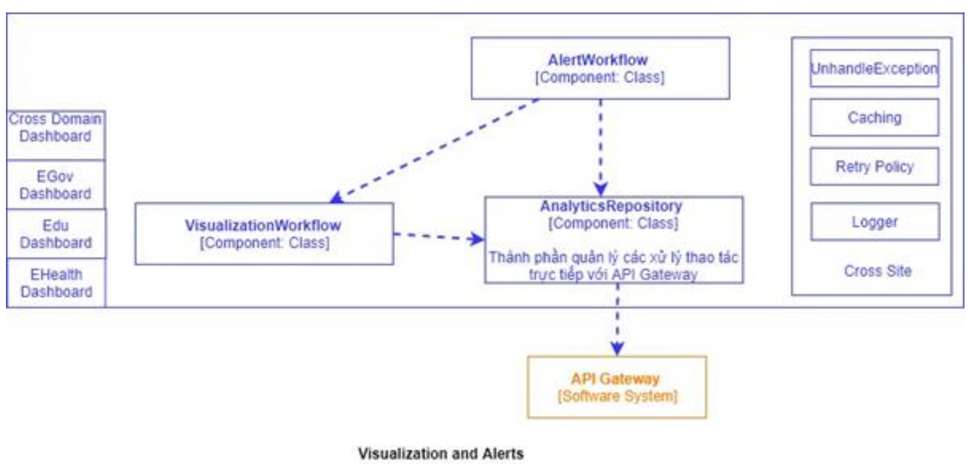

Fig 2.2.2-11: Visualization and alerts

Components to build Visualization and alerts:

- AlertWorkflow: the main component responsible for warning when the analytical data shows signs of deviation from the set KPI or the data has reached the threshold.
- VisualizationWorkflow: the component responsible for drawing graphs and charts to visualize analytical data; AlertWorkflow calls this component.
- AnalyticsRepository: the component responsible for directly interacting with API Gateway to get analytics data.

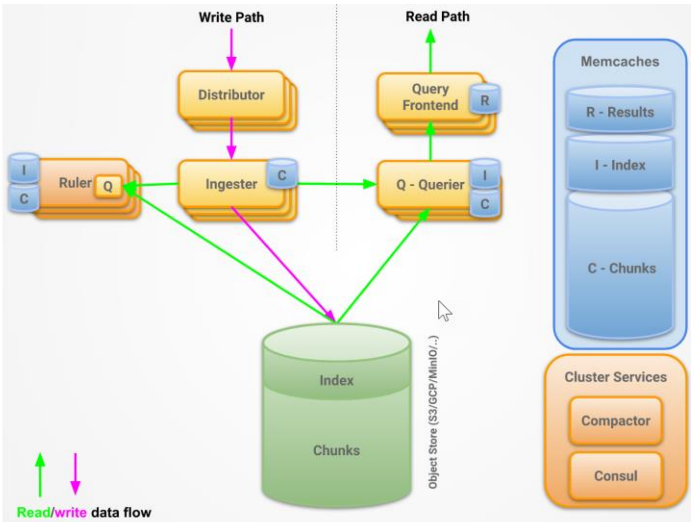

Fig 2.2.2-12: Realtime view

The real-time reporting indicator update module is built to manage all notifications and warnings when problems are detected (triggered by changing KPI status or event correlation).

Data read and write done on two separate channels to increase the system's speed. In which the information is stored mainly in Memcached, 02 main streams include:

+ Read: Query frontend và Querier

+ Write: Distributor và Ingester

2.2.3 Technological components

Here are the technologies to build an IOC system:

- Visualization and Alert: the system uses programming language on Ubuntu operating system to create charts, graphs, and alerts.
- API Gateway: the system uses Netflix Zuul, Netflix Eureka, Netflix Ribbon to form a complete API Gateway with Load balancing Proxy and discov-ery service.
- Analytics Services: this component is built by NodeJS, MongoDB language on the Ubuntu OS platform.
- Analytics Portal Management: the system uses Angular programming language on Ubuntu operating system to create a management tool for data models, graphs, charts, and KPIs.
- SQL Data Warehouse: This component is built by MySQL database or Oracle Database, depending on the project.
- Streaming Analytics: this component uses Pentaho Kettle to analyze Stream.
- Batch & Stream Processing: the system uses Pentaho Kettle to process Stream and raw data.

– Raw Data Store: use MongoDB to store data after entering the hub.
– Streaming Hub: use the Kafka cluster to collect streaming data.
– Bulk Loading Hub: uses Pentaho DI to collect transaction data..

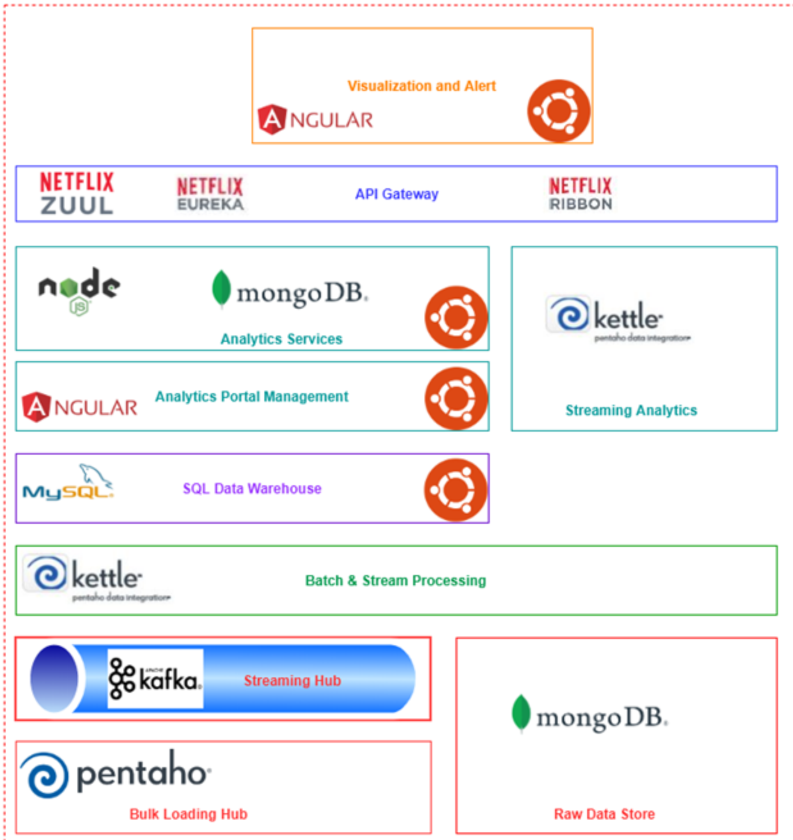

Fig 2.2.3: Technology used

Chapter 3 Results

3.1 System implementation

3.1.1 System deployment

In the previous program, a document has presented the thesis about the Data Integration module and how to design the system's Data Warehouse. At this item, the text will display the way the data from other systems to Data Warehouse

3.1.1.1 Data Integration

Data integration, the thesis is divided into two parts (first run, Schedule)

- The First run: will integrate all data before the current time
- Schedule: will incorporate new data that arises and changes over time

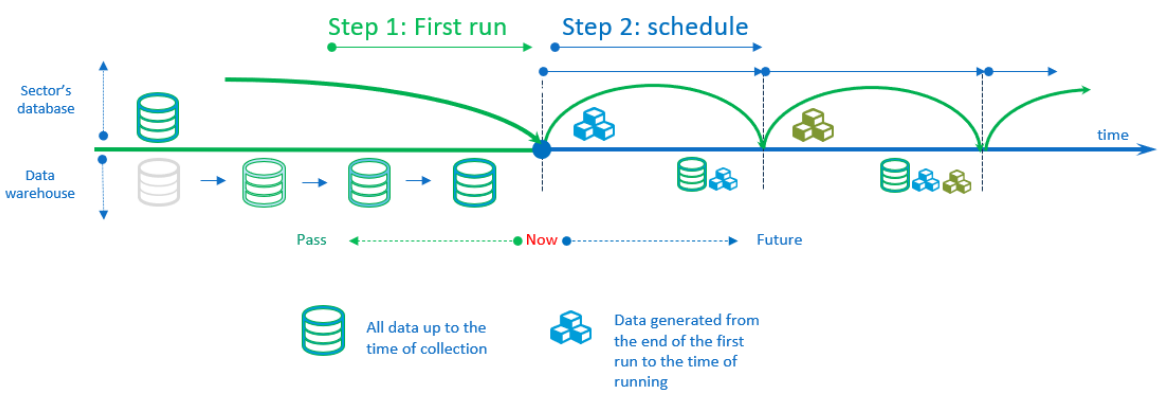

Fig 3.1.1-1: Data Integration Design Process

The design process of a fundamental transformation includes the functional components of Pentaho DI:

- Table input: SQL statement to get API source data declared in DB
- REST Client: configure information, API methods (get/post),...
- JSON input: configure the information returned by the API fields
- Select values: filter the fields needed to ETL data at the Staging level, convert, process data types, rename fields, etc.
- Filter rows: process, filter the value of data fields (duplicate, business data, ..)
- Table Output: Table stores data; in this step, you can check if the table exists in the database; if not, you can run the SQL command to create a table based on the column fields that have been filtered out in the selection step. Values. The detailed process is shown in Figure

below.

Fig 3.1.1-2 : Transformation ETL (first run)

Similar to designing transformation for scheduling, adding data processing steps (Execute SQL Script)

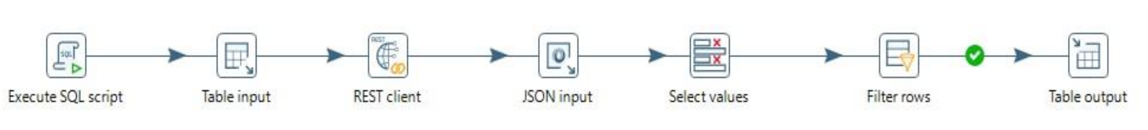

Fig 3.1.1-3: Transformation ETL Data (schedule)

Design Job ETL with medical data (medical examination and treatment), including transform-nations that perform ETL according to the corresponding input data source, and design an additional transformation to record the job execution. monitoring and monitoring system for ETL jobs data

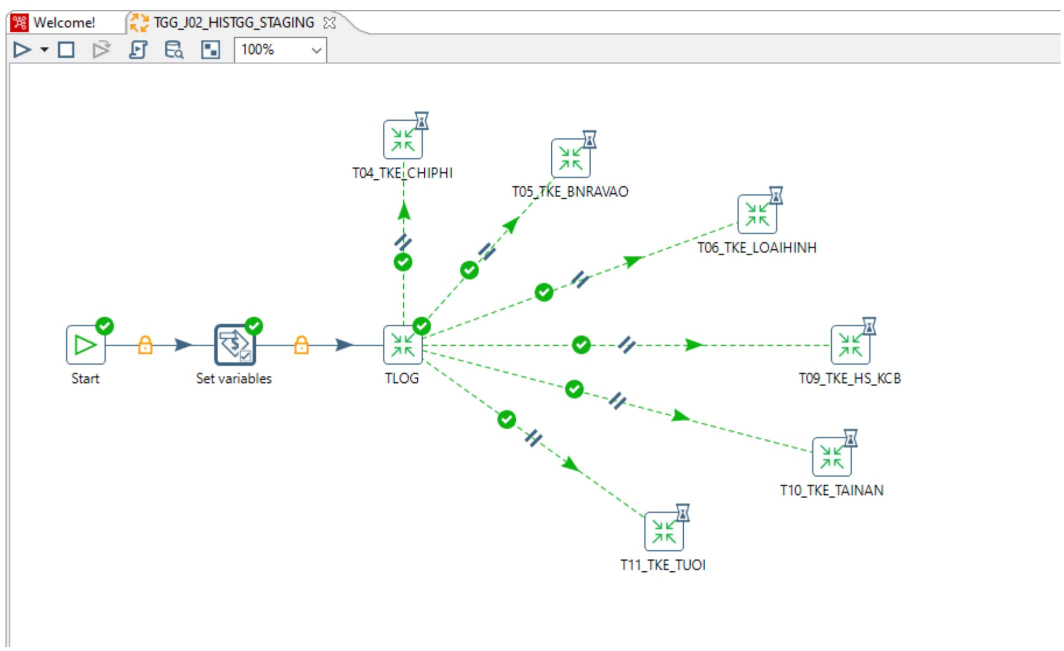

Fig 3.1.1-4: The Job ETL data in eHealth (Staging)

Design Job ETL data of the field of aggregation at the Staging layer & process the transformation to the Data Mart layer

Fig 3.1.1-5: Job ETL a health's data

3.1.1.2 Deploy to the system

Process explanation:

- Login: Log in to the system through the account (username) and password provided (password).
- Logout system: perform log out from the system
- User management, decentralization: Add, update, delete permission groups/users
- Update Transformation/Job: Add new, edit, delete Transformation load input data from Excel, API, Database Connection, Kafka with Spoon (Pentaho DI) pushing to Staging layer
- Adjust the execution schedule of Transformation/ Job: Add, edit, delete the execution schedule of Transformation
- Monitor and monitor Job/transform execution: Build, configure Job to check whether jobs execute at a certain time & error warning (if any)
- Detailed steps and illustrations are presented in the appendix of the thesis

No	Feature	Description
1	Login	1. Login
2	Logout	1. Logout
3	User managerment	1. End-user can create a new user 2. End-user can edit user's info 3. End-user can delete user
4	Role management	1. Enduser can create a new role 2. Enduser can check and change a role 3. Enduser can edit a role

		4. Enduser can delete a role
5	To create Tranformation with data input from file Excel,CSV,.. by sector in Staging	1. Enduser can create a tranformation file with: + Microsoft Excel Input + CSV file input 2. Enduser can filter a filed what need to ETL on Staging 3. Enduser chooses a datasoure in Staging 4. The user checks if the Table already exists in the database in Staging: + If not: excute a SQL what create a table, and come back Step 5 + If any: next to new step
6	Create Transformation with input from each field API at Staging	1. User creates a Transformation (.ktr) file with an Input option containing a list of APIs by field to ETL data: + Table input + API (JSON input) 2. User list & filter a field which required to ETL Staging Level Data 3. User selects DataSource to store data at the Staging level 4. The user checks if the Table already exists in the database in Staging: + Not yet: execute the command to create Table -> Go to step 5 + Already available: go to step 5 5. User selects Table to store data & conduct data ETL at Staging level
7	Create Transformation with Database Connection input data by the field at the Staging level	1. User creates a Transformation file (.ktr) with the input option containing data information for each field that needs ETL as table input 2. User lists & filters fields required to ETL data at the Staging level

		3. User selects DataSource to store data at the Staging level 4. The user checks if the Table already exists in the database in Staging: + Not yet: execute the command to create a table -> Go to step 5 + Already available: go to step 5 5. User selects Table to store data & conduct data ETL at Staging level
8	Create a Job to execute Transformations with Spoon	1. User creates Transformation files (.ktr) to ETL Staging level data 2. User creates Job file (.kjb) to execute transitions at the same time and schedule how often to execute one time 2.1 The user selects transformation in turn in the order of execution 2.2 User Timer Job Execution
9	Upload the file Transformation/Job designed with Spoon to Pentaho Server	1. User login to the system (user/pass) 2. Select the file (.ktr/.kjb) to upload to the Server 3. View the list of files that have been uploaded to the Pentaho Server
10	Delete the file Transformation/Job designed with Spoon to Pentaho Server	1. User login to the system (user/pass) 2. Select the file (.ktr/.kjb) to delete in the Server 3. View the list of files that have been deleted to the Pentaho Server
11	Create a schedule to execute Transformations with Spoon	1. User login to the system (user/pass) 2. Select the file (.ktr/ .kjb) to configure, schedule the execution of data ETL how often once 3. View the list of schedules that have been executed

12	Scheduling Job/Transformation execution lock with PIC	1. User login to the system (user/pass) 2. Create a schedule that locks the execution of a transformation (.ktr), Job (.kjb) cannot execute for some time 3. View the list of locked scheduled schedules
13	Update execution schedule of Transformations with Pentaho User Console (PUC)	1. User login to the system (user/pass) 2. Select the schedule to be updated and proceed to adjust and update the information 3. View the list of scheduled/adjusted schedules
14	Deleting the execution schedule of Transformations with Pentaho User Console (PUC)	1. User login to the system (user/pass) 2. Select the schedule to delete and delete the corresponding schedule 3. View the list of schedules that have been executed
15	Pause the execution schedule of Transformations with Pentaho User Console (PIC)	1. User login to the system (user/pass) 2. Select the schedule to be paused and execute the corresponding schedule pause 3. View the list of schedules that have been executed
16	Update Job/Transformation execution lock schedule with PUC	1. User login to the system (user/pass) 2. Select the schedule to be updated and proceed to adjust and update the information 3. View the list of locked scheduled schedules
17	Assigning Transformation/Job to the schedule locks Job/Transformation execution with PUC.	1. User login to the system (user/pass) 2. Create a schedule that locks the execution of a transformation (.ktr), Job (.kjb) cannot execute for some time 3. Select the file .ktr/.kjb and select the corresponding lock schedule 4. View the list of locked scheduled schedules

18	Delete Job/Transformation execution lock schedule with PIC	1. User login to the system (user/pass) 2. Select the schedule to delete and delete the corresponding schedule 3. View the list of locked scheduled schedules
19	Monitoring of Transformation/Jobs that have been scheduled to execute periodically	1. User login to the system (user/pass) 2. View a list of transformations/ jobs that execute periodically 3. View detailed information of each transformation/job: canvas preview, log execution

Table : The data integration’s feature

3.1.1.3 Dashboard design

Based on KPIs in each field, the system will conduct detailed analysis and data visualization to help leaders overview the industry's data.

❖ Dashboard analyzes data according to KPIs in each field.

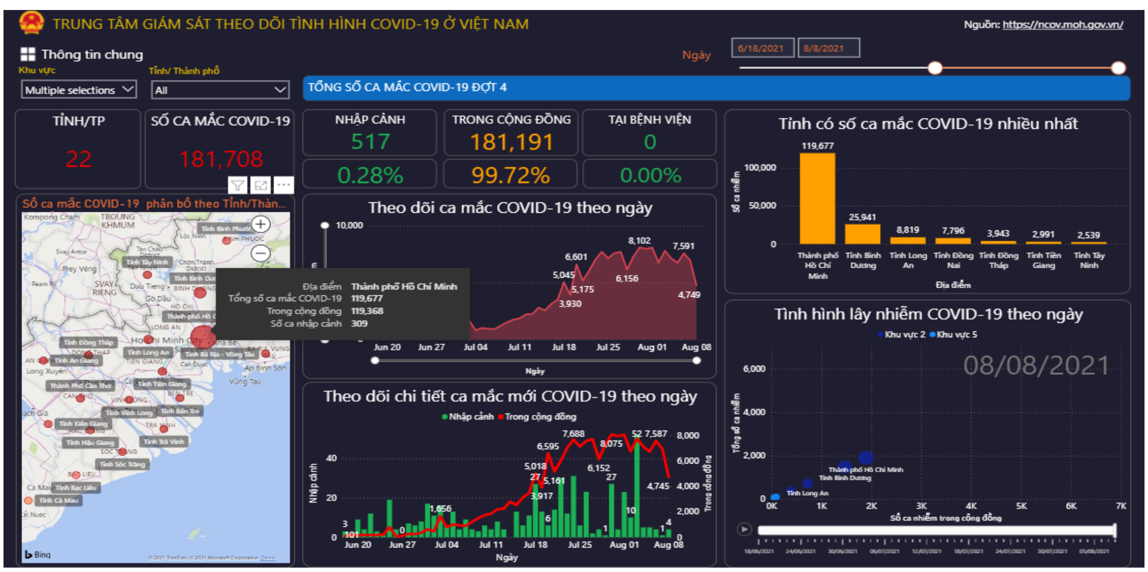

Fig 3.1.1-1: Sector’s visualization

In addition to building a dashboard system for the integrated areas, the system also monitors itself to ensure efficient operation.

The data integration system is built with the Cluster model, the monitoring of job execution is to address the following specific goals:

- Configure monitoring, tracking which jobs have been executed on which Pentaho server at a particular time in the Pentaho Server Cluster
- Visualize the above data to monitor and monitor detailed systems in servers for data ETL
- Shorten tracking time, monitoring Job ETL data

Before expanding the monitoring module, monitoring the data ETL system, the monitoring implementation is based on PUC (shown above).

- Steps to monitor and monitor the data ETL system:
- Configure Job for centralized logging
- Analyze logs and conduct Visualization of those data

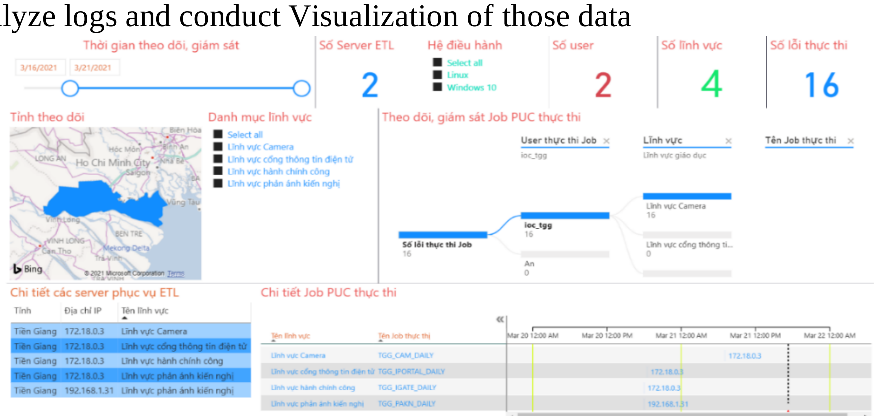

Fig 3.1.1-2: Log analytics

Monitoring parameters in the model include:

- Monitor the number of servers serving ETL

- Monitor the number of fields that are executing Job on the system
- Detailed statistics of servers serving the ETL system
- Query time monitoring for the entire system
- Monitor the number of Job execution errors (if any)

Features include:

- All this information are multi-dimensional filters; when adjusting information in one chart, the rest of the graphs in the system will be updated according to
- Export data,...
- Also, configure alerts via Telegram when the job/Transform executes with errors

3.1.2 System implementation

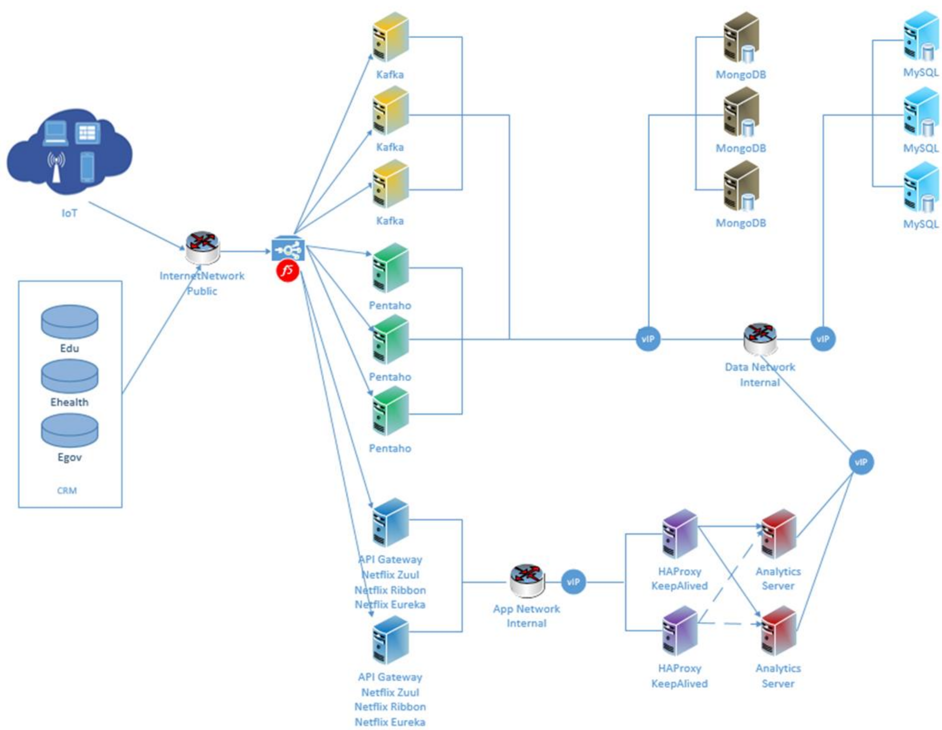

Fig 3.1.2-1: System implementation

TT	Loại server	Chức năng
1	Kafka	System queue
2	Data Integration	To integration a data
3	API Gateway	System's API Gateway

4	MongoDB	To save unstruct data
5	MySQL	To save struct data

Table 3.1.2-1: IOC’s component

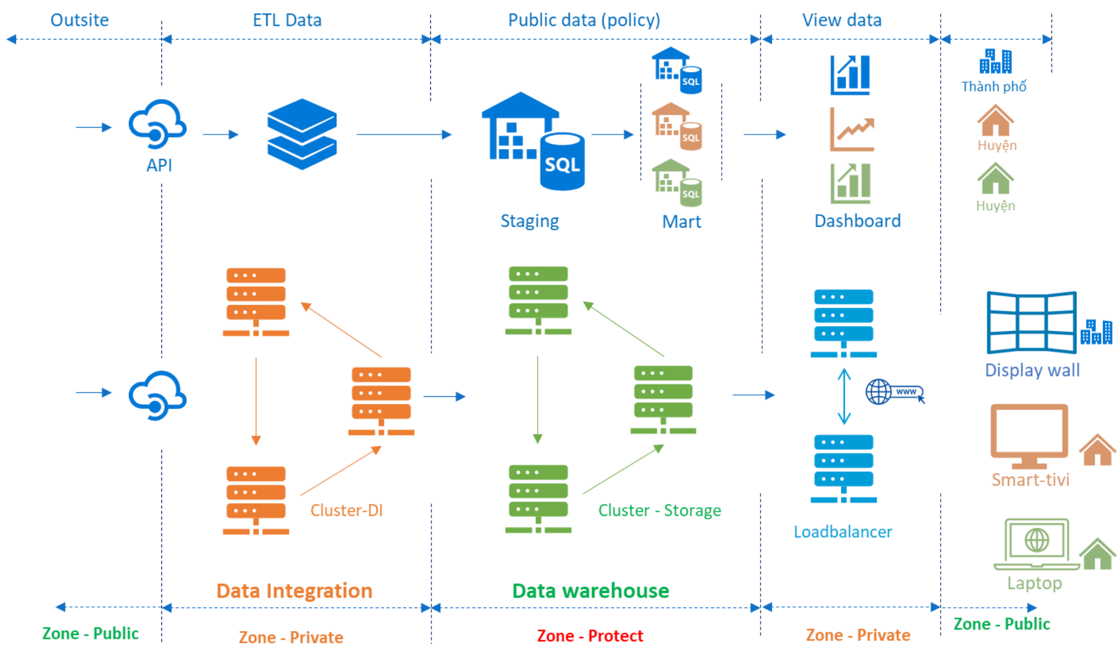

Fig 3.1.2-2: IOC’s Zone

No	Zone	Feature
1	Cluster-DI	Groups of servers are installed in a cluster model. This group of servers is located in the Private zone, used for data integration. Due to security factors, the server group is connected via intranet range or VPN
2	Cluster – Storage	The group of servers is installed to become the system's database. This group is installed in its strong partition, accessed only with the Data Integration and Application layers
3	Application	A group of servers is installed for building applications in the system.

Table 3.1.2-2: IOC's cluster

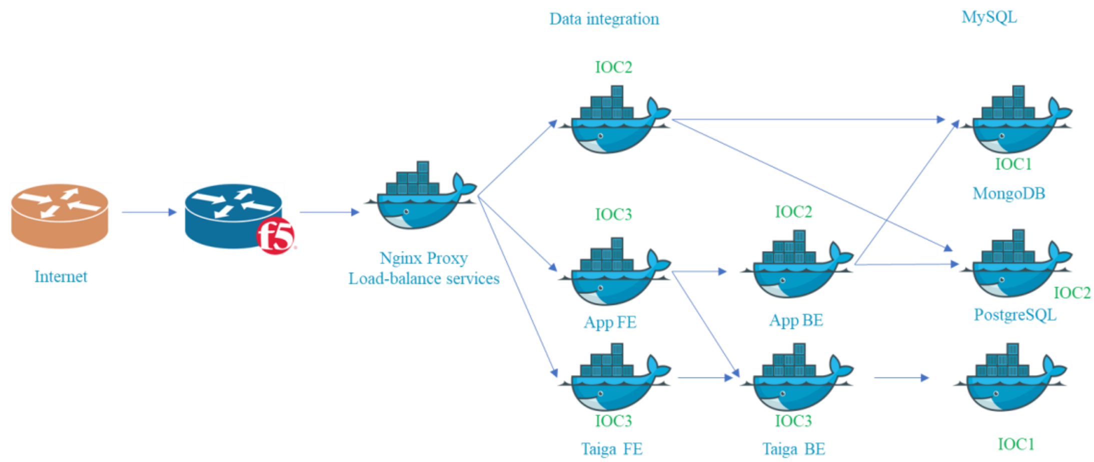

Fig 3.1.2-3: Docker Swarm Container

No	Container	Feature
1	Nginx proxy	This is load balance modul of system
2	IOC1: MongoDB	This is a modul what saves unstruct data
3	IOC2	This has a backend of system and PostgreSQL database. PostgreSQL database is a repository for Data interation model
4	IOC3	This has a backend, frontend of opetation system in IOC.

Table 3.1.2-3: IOC's container

3.2 Result

With the integration and visual presentation of information and data from different fields, IOC has provided the city authorities and the government with a comprehensive overview of all socio-economic activities. From that, they can make timely and accurate decisions, saving a lot of time and effort in the process of monitoring and operating the work. Through the IOC system, the interaction between citizens and the government is also improved, contributing to improving the level of satisfaction of citizens and businesses with the operation of the government.

The IOC enhances the citizens' experience by making infrastructure navigation simpler. In the cities and provinces that have deployed the IOC, the quality-of-life indicators have significantly improved. As shown in Figure 3, Binh Phuoc operated a comprehensive IOC system with 10 services in 2020. After more than 2 months of trial operation, the authorities received over 150 complaints, petitions, calls and messages daily from citizens in the security, fire prevention, traffic accidents, public administration sectors. The rate of receiving and handling level 3 and level 4 online public services has been increased from 9% at the beginning of the year to over 90% in September.

Fig. 3.2-1: VNPT's technical team works 24/7 to support the operation of Binh Phuoc IOC

Fig. 3.2-2:. Da Lat Intelligent Operations Center

Officially operated in December 2019, Da Lat IOC showed significant results as shown in Figure 4. The citizens' feedback to the city agencies were processed slowly from a few days to a few months, it takes about 2 to 24 hours from receiving complaints to complete processing in some cases. After operating for 6 months, the Da Lat authorities received an average of 30 to 40 complaints every day in different sectors [19]. Dalat.vn tourism portal and smart travel application have connected with the IOC to update and track data from the system. Visitors can search and update information about the city, book tickets, hotel rooms online or look up cultural events, food, traffic and weather easily.

The IOC also provides the city agencies and the government with a comprehensive status of the city activities, which increases the effectiveness of public services and city safety. A positive result of the Binh Phuoc IOC is through the traffic safety monitoring and operating system integrated with traffic cameras, about 15,000 traffic violations have been recorded with 13 billion VND fine. The camera system has helped detect many cases of security and traffic violations, importantly contributing to ensure public security in the province. The government and city agencies can update the development in the socio-economic situation and other problems in various fields, which helps them to forecast, assess the situation then accurately handle the problems.

Economic developments have been demonstrated in the implementation of the IOC system. The National Document Communication Axis has connected with 100% of ministries, city agencies and organizations with over 2.2 million electronic documents sent and received that saves the national budget more than 1,200 billion VND per year [20]. The National Public Service Portal opened in December 2019, after 8 months of operation, it has connected with 18 ministries, agencies, 63/63 cities and provinces, with nearly 56,4 million accesses, more than 220,000 registered accounts, over 14 million synchronized records. The total costs saved when operating public services are estimated at more than 13,000 billion VND per year, of which, the National Public Service Portal contributes over 6,700 billion VND per year [20].

3.3 Conclusion

Smart city is the development objective of many countries in the world including Vietnam, with the construction of IOC as a smart brain that collects and operates the cities' activities by collecting and standardizing data. This thesis proposes an IOC architecture for smart cities in Vietnam with five main layers which includes data integration, storage, data analytics, data visualization and application layer. Acting as the smart brain of a smart city, the IOC helps local organizations and authorities in integrating data from many systems to create an interactive, connected platform that provides a comprehensive overview of a city' activities. The results show that this architecture increases the efficiency of the decision-making process by reducing the time and enhances the citizens' experience. Future work is planned to optimize the system process and implement Artificial Intelligence models in the application layer.

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Appendices

1. IOC benefits in the model smart city versus media

1.1 For people and businesses

The benefits of smart cities for citizens and businesses can be outlined through three general activities[1]:

- Improve the quality of life: Smart cities ensure a comfortable, positive, healthy, and safe living environment. Residents can enjoy utilities including low-cost energy use; convenient public transportation system; students can study in good quality schools; emergency response units respond quickly and promptly; fresh air, clean water source; low crime rate; and various entertainment activities.
- Improve working efficiency: Employees are provided with basic infrastructure services to ensure good competitiveness in the world market. These infrastructure services include: Broadband Internet connection; clean, stable energy sources at low cost; opportunities to learn, improve skills and knowledge; the cost of living and working space is affordable; and efficient transportation system.
- Sustainable development: Smart city helps to use resources and save costs effectively. Thereby, taxpayers' money is used rationally to improve public services' quality further. The efficient use of resources also ensures benefits for future generations.

1.2 For urban governance

One of the biggest problems facing cities today is the traditional urban governance model (Figure PL1.1). The conventional governance model is built around service providers, operates independently, and developed along vertical value chains. In this model, people have to interact with each field by themselves, and unshared data information limits the ability to coordinate and cooperate between sectors and between government and society, creating a cumbersome and slow system that is difficult to change. In reality, when cities expand in size, creating a management structure with high complexity and interdependence between sectors, this model no longer works.

[1] "Smart Cities Readiness Guide". 24/08/2015. Smart Cities Council. Trang 19.

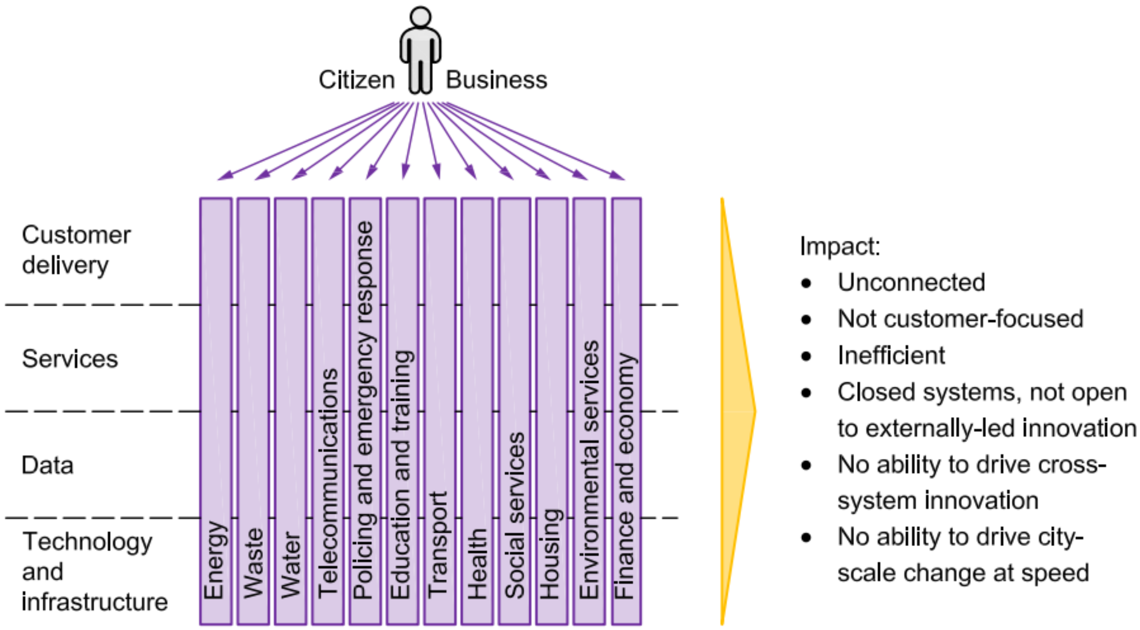
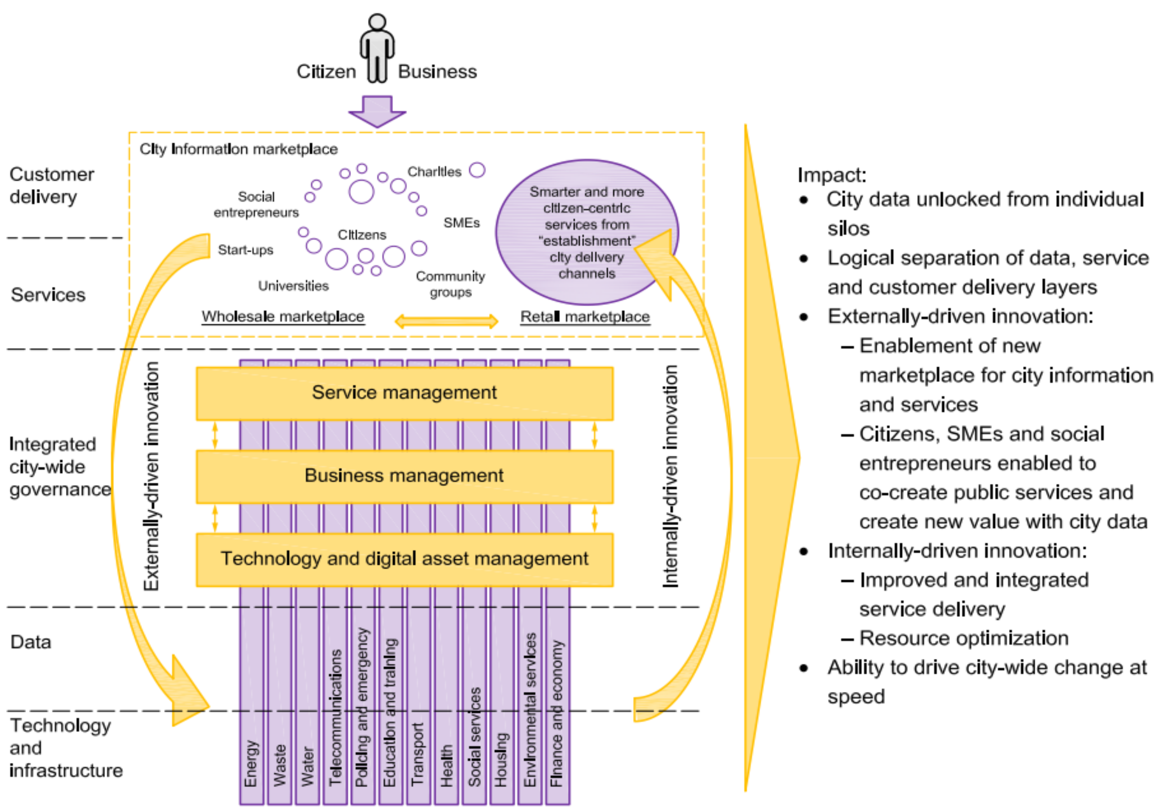


Fig PL1.1 Traditional operating model: where cities have come from [2]



[2] PAS 181:2014 "Smart city framework – Guide to establishing strategies for smart cities and communities". 2014. British Standards Insitution. Trang 14.

Fig PL1.2 New integrated operating model: where smart cities are moving to [3]

With a modern integrated management model towards smart cities (Figure PL1.2), cities can now provide the public with real-time data sources on an open and transparent platform. Interoperability, allowing the integration of services and optimization of city resources. Data (including open city and business data that Data can freely share) will become an asset used to encourage innovative activities driven by the needs of citizens and businesses. Industry and the internal needs of service providers (including government agencies and enterprises), thereby helping to improve the quality of existing services and contributing to the creation of new services. New services and values. City leaders can also balance their budgets more holistically and flexibly to focus on common economic values instead of confined to a particular area. This model also allows establishing a cross-cutting management system to effectively support and evaluate changes at the macro-level. Table PL1.1 illustrates some specific benefits from urban governance under the innovative city-oriented model compared with the traditional model.

	The traditional urban governance model	**Management model in the direction of integration: the destination of smart cities**
Zoning	• Arbitrary and scattered • No cost savings • Limited expansion investment capacity	• Comprehensive and oriented • Resource sharing • Cost savings • Ability to expand investment • Improved planning and forecasting capabilities
Infarstructure	• Inefficient operation • Resource-intensive and expensive to operate	• Optimized by advanced technologies • Saving resources and costs • Improve commitments on service quality • Building on open platforms
Operation system	• Only conjecture about the state of infrastructure • Passive when the problem occurs • Unable to deploy resources effectively to	• Real-time infrastructure status capture • Anticipate and prevent incidents • Use resources efficiently • Automate maintenance work • Cost savings

[3] PAS 181:2014 "Smart city framework – Guide to establishing strategies for smart cities and communities". 2014. British Standards Insitution. Trang 15.

	solve the problem	
Technology investment	• Scattered and isolated • Not optimal in terms of benefits • Unable to take advantage of scale when making significant investments	• Centralized planning • Deployment across management agencies and between projects • Optimizing the benefits brought • Maximum value and cost savings
Participation of the people	• Online connection channels to people are minimal and scattered • People cannot use (or do not readily access) public services in the best way	• A complete interface channel serving both the majority and the minority • People access and use services easily • Citizens can participate in contributing initiatives to the city • Two-way communication between citizens and authorities • Have personalized services for each resident • Citizens can both contribute and access real-time city data, and build applications using the data
Share data	• Separate departments and functions • Departments rarely share data and collaborate to propose initiatives	• Integrated and shared functions and functions • Data is shared between departments and linked to external data services through open standards • More accurate calculation results • Cost savings

2. Design Datawarehouse for the Medical field

The system will focus on analyzing and visualizing health industry data through KPIs in each field as follows::

No	Sector	Discuss
1	Healthcare	- Administration information at Department of Health: Medical examination and treatment, common ICD disease groups, number of emergency cases, deaths, distribution of visits, type of medical examination and treatment (outpatient/inpatient), rate of cases Profile by gender, medical examination, and treatment costs, statistics by age, accident, and injury, use of technical services with health insurance, etc. - Monitor the Covid-19 situation; Infectious diseases (Based on Circular No. 54/2015/TT-BYT dated December 28, 2015, guiding the regime of information, reporting, and declaration of infectious diseases) + Dangerous contagious diseases must be reported on a case-by-case basis immediately after diagnosis, ensuring no later than 24 hours + Dangerous infectious diseases must be reported on a case-by-case basis within 48 hours of diagnosis + Infectious diseases must report monthly cases, deaths, etc. - Medical management at health care level: number of medical examinations and treatment, records with health insurance/without health insurance, number of emergency cases, deaths, detailed accident and injury situation by district and each medical facility - Medical management by day: analyzing and monitoring the situation of medical examination and treatment throughout the province over the days of the month - Monthly medical management: analyzing and monitoring the situation of medical examination and treatment in the whole province over the months of the year - Analyze and monitor the group of health care providers with the highest number of visits, the group of most common diseases, the group of health care workers receiving the most traffic accidents, etc.
2	Dossiers and	- Executive information about the situation of handling

	administrative procedures	administrative procedures: online and in-person receipts, the status of records: unprocessed, processed, on time, overdue,... - Analysis of groups of units with many overdue and unprocessed records - Analyzing groups of fields with many overdue and unprocessed applications - Follow up the processing of overdue dossiers, handle dossiers by field
3	Text, executive	- General statistics on document management situation: coming, going by unit - Statistics on approval status, document processing, system usage (frequently or rarely using system access, ...) - Analysis of word processing performance: groups of units have many documents, browse many documents, many documents have not been processed, units rarely use the system, ...
4	Monitor and monitor system monitoring	Monitor scheduled execution of ETL system configured in Pentaho Server
5	Monitor the situation of the COVID-19 epidemic	Monitoring and analyzing the ongoing situation of the COVID-19 epidemic (phase 4) in Vietnam, data source from the Ministry of Health

Table 2.1 – Ehealth’s KPI

From the business analysis KPIs for each medical field, students have designed a data warehouse of the healthcare industry according to the star model (Star schema).

2.1 Dimention dtable

TT	Dimension Table	**Discuss**
1	Yte_dm_benhvien	Store information about medical facilities (hospitals, medical centers, polyclinics, medical stations, private clinics, ...)

2	Yte_dm_khoa	Save information about medical facilities
3	Yte_dm_benh_icd	Save information coding diseases, causes of death ICD-10 according to Decision No. 4469/QD-BYT dated October 28, 2020
4	Yte_dm_doituong	Save health insurance subject information
5	Yte_dm_btn	Save information on the list of infectious diseases according to Circular No. 54/2015/TT-BYT dated December 28, 2015
6	dm_coquan_donvi	Store information about agencies, departments, and agencies that receive and handle administrative procedures
7	Dm_linh_vuc	Save detailed information in the fields of document processing and administrative procedures
8	Dm_ky	Save time information according to the hierarchy of days, months, quarters, years

Table 2.2 – List of Dimension Table

2.2 Fact Table

No	Fact Table	**Discuss**
1	Yte_tke_benhvien_ngay	Save detailed information about the number of times in and out of medical facilities for medical examination and treatment by type (outpatient medical assessment, outpatient treatment, inpatient treatment), several emergency cases, deaths, patient records male/female, with health insurance/without health insurance by time dimension, medical facility.
2	Yte_tke_benhvien_thang	Save detailed information about social insurance payment costs by time dimension, medical facilities
3	Yte_tke_theo_doituong	Save detailed information about medical examination and treatment records with health insurance according to the time dimension, medical facilities, and social insurance beneficiaries.
4	Yte_tke_theo_dotuoi	Save detailed information about medical examination and treatment records according to a specific age

		frame by time dimension, medical facility
5	yte_tke_tainan	Save detailed information about accidents and injuries (traffic accidents, labor, burns, ...) in the direction of time, medical facilities.
6	Yte_tke_ca_mac_icd	Save detailed information about the number of cases by time, medical facilities & ICD
7	Yte_tke_dich_benh	Save detailed information about the disease situation by time dimension, medical facilities
8	Yte_tke_benh_truyen_nhiem_24h	Save detailed information about the situation of dangerous infectious diseases, which must be reported immediately after diagnosis, ensuring no later than 24 hours. [4]
9	Yte_tke_benh_truyen_nhiem_48h	Keep detailed information about dangerous infectious disease situations to report within 48-72 hours after diagnosis. [5]
10	Yte_tke_benh_truyen_nhiem_thang	Save detailed information on the situation of infectious diseases must report the number of cases monthly. [6]
11	Igate_tke_theo_coquan	Save detailed information about the processing of administrative documents by the Department of Health
12	Ioffice_tke_xuly_vanban	Save detailed information about the handling of administrative documents of the Department of Health

Table 2.3 – List of Fact Table

[4] Thông tư số 54/2015/TT-BYT ngày 28/12/2015. Phụ lục 1, bảng 1

[5] Thông tư số 54/2015/TT-BYT ngày 28/12/2015. Phụ lục 1, bảng 2

[6] Thông tư số 54/2015/TT-BYT ngày 28/12/2015. Phụ lục 1, bảng 3